GCAM SOFTWARE DOCUMENT SET

GCAM-SW-VVR · Verification and validation report

> Demonstration document — generated from the repository's Markdown; not a regulatory submission.

Document	GCAM-SW-VVR · Verification and validation report
Revision	2026-10-23-1
Date	2026-10-23
Documentation source commit	0e58c93396b3d3ea6453555867800a3bc9e113e8
Source tree	clean at 0e58c93396b3d3ea6453555867800a3bc9e113e8
M4 test subject	84c6ee2d60a7fa06943350290eaef4a173f48eb0
Manual companion deployment	2e7e30ec32aff40a619d92ddd625eeb0c6d6e776e2892ef3360e7b3f4b7359bb
Approvals	not applicable (demonstration)

Document control

Source history endpoints (first / last commit; no invented approval history).

Source	First commit / date	Last commit / date
docs/VV.Studio.Acquisition.md	0ed17e0 2026-10-01	9d55ad1 2026-10-04
docs/VV.Studio.Detector.md	1104324 2026-10-02	e0c36a1 2026-10-02
docs/VV.Studio.Imaging.md	a2657d1 2026-10-02	f4722bf 2026-10-07
docs/VV.Studio.Waveform.md	1c3ae61 2026-10-02	47384c5 2026-10-02
docs/VV.Studio.md	5493cc2 2026-10-01	6516cb7 2026-10-23

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VV.Studio — verification and validation of GCAM Studio

Scope: the three Studio projects (`Gcam.Studio.Core`, `Gcam.Studio.Services`, `Gcam.Studio`) and their tests. Not covered: the simulation engine (verified by its own test project, `tests/Gcam.Tests`; the product-level evidence is VV.Gcam.Evidence) and the original code-behind viewer `src/Gcam.Wpf` (removed in TODO-12 (2026-10-02); last present at `85b2ed1`). This document says **how we know Studio does what it must**; it is one of a set:

Document	IEC 62304 activity	Holds
VV.Gcam.URS · VV.Gcam.PRS	system level, above §5.2	user needs and product requirements for the whole Gcam product concept; Studio is its subsystem SS-4
VV.Studio.SRS (GCAM-SW-SRS)	§5.2 requirements analysis	the requirements (`SR-*`), inputs / outputs, messages, risk-control requirements
VV.Studio.SDS (GCAM-SW-SDS)	§5.3 architecture, §5.4 detailed design	software items and units (`SU-*`), interfaces, SOUP, requirement allocation
VV.Studio (this page)	§5.5–5.7 verification, validation, §7 risk, §9 problems	traceability to tests, validation scenarios, anomalies, risk table

The working design guides are DESIGN.Architecture and the other `DESIGN.*` pages.

At a glance

- 124 active software requirement IDs plus nine withdrawn IDs (§4). Live acquisition verification and measurements are recorded in VV.Studio.Acquisition; desktop acquisition validation passed in the final desktop pass on 2026-10-02.
- Twelve regression scenarios, a plot gate and a 16-frame diagnostic survey are opt-in desktop tests. Scenarios judge product state using independent oracles; every corrupted verdict fails. VAL-09 … VAL-13 below cover Spectrum, Waveform, Detector, isotope imaging and retained-flood focus. Screenshots never decide a numerical verdict.
- The layering that keeps the logic testable is compiler-enforced (SR-ARCH-01, -04).
- Open problems are in §6 (AN-01 … AN-16); no screen-reader session has been run.

Dated implementation records below retain their execution limits and counts. Subsequent analysis identified the Co-60 bias as undersampling; current verification totals are in Current test inventory.

TODO-08 phase B verification (Release, 2026-10-02): the reviewer reported 259 engine, 86 Studio Core, 28 services and 10 UI-oracle tests passing and a build with 0 errors. The measured imaging results, including the Co-60 bias finding TODO-17 with its cause open, are in VV.Studio.Imaging. Local `dotnet build Gcam.sln -c Release -m:1` passed with 0 warnings and 0 errors (the default parallel invocation exited 1 without a diagnostic error). The offscreen render opt-in passed after removing the options panel's Border ancestor binding, regenerating eight mixed-scene PNGs plus the existing eight workspace and four plot PNGs. Local `dotnet test Gcam.sln -c Release -m:1 --no-build` also completed successfully: 259 engine, 86 Studio Core, 28 services and 10 UI-oracle tests passed; nine desktop tests and the independent opt-in render case skipped. A later shell creation failed with CreateProcessAsUserW error 5; no further commands were launched. The final binding-identity assertions compiled in the local build but were added after the passing render run; their rerun and visual inspection were not performed. Desktop validation remains pending, and no Studio launch or desktop UI test was performed.

TODO-09 adds effective editable optics and retained-data refocus. The added deterministic mixed-preset regressions use 20,000 accepted list-mode events per preset/plane, complete isotope lines and activities Cs-137:Co-60:Co-57=1:1:1.5 at normalized positions (-0.5h,-0.3h),(0.5h,-0.3h),(0,0.5h). The gate is one resolution element per isotope channel, with/without stripping; it is not a broadband or sub-mm guarantee.

The sampling regression uses geometric Cs-137, 1 m, 18 mm detector, pitches 0.6/0.3 mm, y=-10..10 in 1 mm steps, 1,000,000 biased histories/position, seed 12345 and 0.25 mm recon grid. It asserts improvement with finer sampling across positions, not a universal threshold. Offscreen images are explicitly analytic drawing fixtures, not MC evidence. Release verification (2026-10-02): `dotnet build Gcam.sln -c Release -m:1` passed (0 errors; one existing xUnit2000 warning). `dotnet test Gcam.sln -c Release -m:1 --no-build` passed: 259 engine, 102 Core, 40 services and 10 UI-oracle cases; nine desktop cases and the opt-in render case skipped. The separately logged sampling regression (started before the command block) also passed: RMS/max 1.0734/1.9966 mm at 0.6 mm pitch and 0.5847/0.9525 mm at 0.3 mm pitch (re-measured 2026-10-02 with the replaced random generator; previously 1.0810/1.9966 and 0.5837/0.9511) under the 1,000,000-history conditions above. This reduced-budget run is distinct from the theme-55 3,000,000-history evidence quoted in the UI. With `GCAM_RENDER_SNAPSHOTS=1`, the render case passed without opening a window, generating 16 additional F=800 expanded/collapsed optics images alongside the refreshed existing captures in `assets/studio-render/`. Desktop validation remains deferred; Studio was not launched. Visual inspection found the preset selection displayed the record's diagnostic string. A final `OpticsPreset.ToString()` override now returns the preset name. Its rebuild command, `dotnet build Gcam.sln -c Release -m:1 --no-restore`, was blocked before process creation with `CreateProcessAsUserW failed: 5 (access denied)`. No further verification commands were launched. The counts above precede this final display-only change. The planner subsequently verified the final Release build (0 errors), all 259/102/40/10 tests and the render case (1/1), and inspected all 16 new optics snapshots, including the derived geometry at F=800.

Long numerical measurements now use `GCAM_EVIDENCE_TESTS=1`; ordinary test runs and CI report them as skipped. The full sampling sweep and the 20-seed precision measurement keep their existing budgets and output. Their evidence and reproduction command are in VV.Studio.Imaging. The separate small off-axis sampling regression remains in the default suite; its seed-spread measurement also uses the evidence opt-in. Follow-up command execution was blocked while reading engine code (`CreateProcessAsUserW failed: 5`, access denied); no further commands were started, so the revised default service-suite duration and final regression verification are unavailable locally. The already-running budget calibration completed successfully: 100,000 histories/position at y={-8,-4,4,8}, five seeds, fine/coarse RMS ratios 0.463150..0.591137. The revised gate is <0.85 (observed max + twice the observed range, rounded up). Full-grid calibration took 193.9 s; the final fast test restricts reconstruction to ±12.125 mm on the same 0.25 mm grid phase. Its margin, seed table and the distinction between measured calibration and unverified final runtime are in VV.Studio.Imaging.

Current test inventory

The generated block below is the current inventory: it names its subject (commit and source snapshot), whether its record is formal or draft, and the selected runner results; it is written by `samples/testing/test_records.py` from retained run records (catalog: VV.Tests (GCAM-SW-STR); milestone reports: VV.Tests.Results (GCAM-SW-STR)). The dated inventory that follows the block is history, not current totals. Desktop evidence is in VV.Studio.History (GCAM-SW-STR); desktop cases not run for the selected subject are reported as not executed.

History (2026-10-04). Verification of the ambient default and the fixed spectrum axis, 2026-10-04, on the tree committed as `13387fb`. Commands: `dotnet build Gcam.sln -c Release` and `dotnet test Gcam.sln -c Release --no-build`. Build: zero errors, zero warnings; the two existing xUnit analyzer warnings (`ImagingServiceTests.cs:106`, `WaveformServiceTests.cs:118`) reappear only when `Gcam.Studio.Services.Tests` is fully recompiled. Normal test execution:

Subject `84c6ee2d60a7fa06943350290eaef4a173f48eb0`; formal; source snapshot `cf007b620be300174f69b44f8f62eb247bbf43f4a9cbd4fb56ba88be0dfb6640`.

Assembly / family	Passed	Failed	Not executed	Missing cases
Gcam.Studio.RenderTests	16	0	15	0
Gcam.Studio.Services.Tests	152	0	7	0
Gcam.Studio.Tests	309	0	0	0

Assembly / family	Passed	Failed	Not executed	Missing cases
Gcam.Studio.UiTests	297	0	4	0
Gcam.Tests	595	0	0	0
checker	7	0	0	0
cocotb	111	0	26	0
unittest	284	0	1	0

No desktop, render or long-evidence opt-in was enabled for this run. The retained final desktop record reports 27/27 total UI cases (13 headless oracles + 12 scenarios + gate + survey); **27 is a suite total, not a scenario count**. The retained RTL execution record in `rtl/README.md` reports 137 cocotb cases with C# vectors (26 configurations, 452,608 exact sample comparisons). Neither desktop nor cocotb was rerun for this record. **Desktop pending:** the desktop scenarios have not been run since Studio starts with the ambient field on (SR-RUN-29) and draws the fixed 0–2000 keV spectrum axis (SR-SPEC-01); no desktop scenario covers the ambient panel or the dose warning yet. Dated inventories and execution restrictions below remain history; use this section for current normal-run totals.

Verification after stage 2 (2026-10-13, not a milestone record). After declutter stage 2 and the desktop-review redesign (TODO-44 / 45, tree `0d21524`): Release build 0 warnings, normal suite **1,010 passed / 48 skipped**, opt-in offscreen renders 25 / 25, catalog 193 units, CI checkers pass (History (GCAM-SW-STR, Declutter stage 2 and desktop-review redesign, 2026-10-13)). **Desktop pending:** the desktop scenarios were **not** re-run after stage 2 (the desktop was not available). The stage-2 desktop run — the whole desktop suite plus broken-verdict / recovery runs for the changed assertions, and the UI-automation re-measure and re-pilot including the no-title-bar assertion — waits for the author's go; the desktop statuses dated 2026-10-12 or earlier in the matrix below predate stage 2.

Latest verification (2026-10-13, after the access-key / Options-Enter / cleanup pass; not a milestone record). Release build 0 warnings, normal suite **1,013 passed / 48 skipped** (three new cases), opt-in offscreen renders 27 / 27, catalog 193 units with current discovery. Desktop not run: real Alt-key routing and Enter in the Options dialog on the window wait for the stage-2 desktop run.

Desktop verification (2026-10-13, `ed1298b`). The stage-2 desktop run and TODO-42's scenarios: 28 of 29 desktop scenarios passed, including the no-title-bar scenario and the pilot; the one failure exposed a product defect in the modal focus boundary (AN-17), fixed and re-run with broken-verdict / recovery pairs (History (GCAM-SW-STR, Desktop verification — declutter stage 2 and TODO-42, 2026-10-13)). The "Desktop pending" note above is superseded; real Alt-key routing and Enter in the Options dialog remain headless-only.

Studio manual, first edition (2026-10-15, TODO-43 branch at `51fbdb2`; not a milestone record). Release build 0 warnings, normal suite **1,095 passed / 56 skipped** (new: `ManualRecordTests`, 6 headless; `ManualScenarioTests`, 6 desktop scenarios, opt-in), catalog 220 units, manual generator self-test 24 / 24, Python tests pass. Desktop: manual run 04 passed 6 / 6 and the edition `2026-10-15-1` was generated from it (History (GCAM-SW-STR, Studio manual, first edition, 2026-10-15)).

Studio manual, second edition, and desktop-test videos (2026-10-16, TODO-48 branch at `46c9ada`; not a milestone record). Release build 0 warnings, normal suite **1,171 passed / 63 skipped**, catalog 238 units, manual generator self-test 44. Desktop: the pilot and the observed manual batch on `46c9ada` passed (8 / 8 chapters, including 3.7 export and 4.4 Compare) and edition `2026-10-16-1` was generated from that batch; every recorded desktop test passed with a video (History (GCAM-SW-STR, Studio manual, second edition, and desktop-test videos (TODO-48), 2026-10-16)).

1. Scope and intended use

Intended use. GCAM Studio is a portfolio demo viewer for Gcam, a Monte Carlo simulator of a coded-aperture gamma camera. A user places gamma sources in a scene, runs the simulation, inspects the

detector flood map and the decoded reconstruction, and measures on both images in mm. Users are engineers and reviewers reading the code; the output informs no decision about people or real sources.

Relationship to IEC 62304. This document is structured after IEC 62304 practices — software requirements (§5.2), unit, integration and system verification (§5.5–5.7), risk control (§7) and problem resolution (§9) — because that is the discipline the author works in. **GCAM Studio is not a medical device and not a regulated product. No compliance or certification is claimed.** The safety class below is illustrative.

Illustrative software safety class: B. *If* the same viewer displayed real localisation data that an operator acted on, a wrong coordinate, a wrong ROI sum, or an old image shown as current could mislead that operator. The viewer does not control anything, so the worst plausible outcome is a wrong reading that a person then checks — non-serious injury at most, hence B rather than C. As actually used (a demo on simulated data) it would be Class A.

Engine boundary. Physics, decoding and the scene → config builder belong to the engine. Studio relies on them through Core acquisition, spectrum, imaging, waveform, detector-face and focus-sweep service contracts. The engine inventory is recorded below; `SpectrumServiceTests` additionally verifies the measured spectrum against real-engine deposits.

2. Software items

Items SI-1 … SI-4 and their units SU-01 … SU-26 are defined (Waveform and Detector/focus designs extend them in the later sections) in VV.Studio.SDS §2 (GCAM-SW-SDS, 2. Software items and units (§5.3.1, §5.4.1)). How each item is verified:

Item	Project	Verified by
SI-1 Presentation logic	`Gcam.Studio.Core` (net9.0, no WPF)	unit tests with fakes
SI-2 View geometry and measurement maths	`Gcam.Studio.Core/Imaging`	unit tests, pure maths
SI-3 Simulation adapter	`Gcam.Studio.Services`	integration tests against the real engine (`AcquisitionServiceTests`) + engine-level tests
SI-4 WPF shell	`Gcam.Studio` (net9.0-windows)	inspection + manual UI Automation
SOUP (VV.Studio.SDS §4 (GCAM-SW-SDS, 4. SOUP and required platform (§5.3.3, §5.3.4)))	NuGet	used as published; not separately verified

Only SI-1 and SI-2 can be unit-tested with fakes, by design: everything with logic lives in a project that has no UI stack. SI-3 is tested through the real engine.

3. Software requirements

The requirements are specified in VV.Studio.SRS (GCAM-SW-SRS) — **124 active** in eighteen groups, including navigation, plotting and acquisition. Seven withdrawn batch rows retain their IDs. The matrix cites IDs only; the SRS is the single source of their wording.

4. Verification

Levels follow IEC 62304: **unit** (§5.5) = one class in isolation; **integration** (§5.6) = ViewModels together, or Studio against the real engine; **system** (§5.7) = the running app. Methods: **T** automated test (exact `Class.Method`), **C** compiler-enforced, **I** inspection of code / XAML, **M** manual check driven through UI Automation. Status: **pass** (evidence on 2026-10-01), **partial**, **open** (no evidence yet).

Traceability matrix

Req	Level	Method	Evidence	Status
SR-OPT-01, -02	unit + offscreen	T, I	`OpticsPolicyTests.Presets_ApplyAtomically_MatchPhysicalFieldsAndShowCustom`, `CustomMode_ChangesOnlyThroughSelector_PresetFieldsCannotBeEdited`, `PresetWritesItsValuesAndDisablesFields_CustomEnablesThemFromCurrentValues` (always-shown fields, preset fills and disables, Custom enables from the current values); `OpticsViewModelTests.Focus_AcquiringStoppedCompleted_KeepsEventsAndSpectrum_OpticsLockedWithData`; Optics tab bindings (`IsCustom`, `IsEditable`) inspected in the renders	pass headless + offscreen; desktop re-run after stage 2 pending
SR-OPT-03	unit + integration	T	`OpticsPolicyTests` finite/text/cross-field cases; `OpticsViewModelTests.InvalidInput_StartSelectsTheInputTab_AndDoesNotStartTransport` (Start selects the Optics tab, states the reason in the status bar and starts no transport); `StartRefusalTests.OpticsError_ShowsReason_SelectsOpticsTab`, `SceneError_SelectsSceneTab`, `FocusError_ShowsReason_OpensTheFocalSection`; service config validation	pass headless; desktop deferred
SR-OPT-04	unit + integration	T	`OpticsProjectionTests.Refocus_ReprojectsAllChannels_WithoutMeasurementCalibrationOrNewEvents`, `EmptySnapshot_RefocusKeepsEmptyImagesAndDoesNotCalibrate`; `OpticsViewModelTests` lifecycle/revision/acquisition tests	pass headless; desktop deferred
SR-OPT-05	unit + integration + offscreen	T, I	`OpticsPolicyTests.Geometry_UsesEffectiveFieldsAndHasNoPerformanceBand`; `OpticsProjectionTests.Sampling_SmallOffAxisSample_FinerPixelsReduceLocalizationError` (default); full position sweep (GCAM_EVIDENCE_TESTS=1); sampling evidence; one-line caption with the figures as tooltip / help text (`Imaging.SamplingEvidence`) inspected in the renders	prior full sweep passed; revised default regression verification pending

Req	Level	Method	Evidence	Status
SR-OPT-06	offscreen	T, I	`PlotViewRenderTests.MainWindow_Workspaces_Dark_RenderWithoutWindow` / `_Light_` (three left tabs, the selected tab without scrollable overflow in every workspace at both sizes, scroll gutters), `MainWindow_FocalPlaneAndMlem_Dark_RenderWithoutWindow` / `_Light_` (focal plane F=800), `Shell_OptionsCaptionStatusLeftTabs_Dark_RenderWithoutWindow` / `_Light_` (each left tab and its section group, `Optics.Section` / `Detector.Section`, both themes, 1280×800 / 1440×900); IDs Optics., *Readout.*, Imaging.FocalPlane/Geometry/FocusNote, Imaging.FocalSection/ChannelSection/SweepSection; Measurements under the images inspected	pass offscreen; desktop re-run after stage 2 pending
SR-IMG-01	unit + integration	T	`ImagingWorkspaceTests.SharedWindow_SelectorStripAndRoi_ReuseFrozenAcquisition`; `ImagingServiceTests.WindowChange_ReplaysRetainedEvents_RecalibratesAndMatchesFreshProcessing`	pass (headless); desktop pending
SR-IMG-02	integration	T, I	`ImagingServiceTests`; shared `SpectrumService.BuildBands` and `MeasurementStage`; imaging evidence	pass (headless); desktop pass 2026-10-07 (History (GCAM-SW-STR, Desktop verification — 2026-10-07)): `Imaging_ChannelAndStrip_MatchRetainedFloodAndFoundPeaks`
SR-IMG-03	unit + offscreen	T	`ImagingWorkspaceTests.SharedWindow_SelectorStripAndRoi_ReuseFrozenAcquisition`; two-isotope All / Cs-137 offscreen renders	pass (headless + offscreen); desktop pass 2026-10-07 (History (GCAM-SW-STR, Desktop verification — 2026-10-07)): channel selector in `Imaging_ChannelAndStrip_…`
SR-IMG-04	unit + integration + offscreen	T, I	`ImagingServiceTests.Channels_OffAxisCsAndCo_LocalizeAtTheirOwnSource`, `Precision_TwentySeeds_MeasuresMixedIsotopesAndCoOnlyControl`; `OverlayLabelLayoutTests` (6 cases: gaps, marker/measurement clearance, bounds, recalculation, deterministic packing, omission and invalid geometry); crowded equal-Y and edge-chip renders with diamond overlay and coordinate list	pass (headless + offscreen); measured Co-60 bias in imaging evidence; desktop pass 2026-10-07 (History (GCAM-SW-STR, Desktop verification — 2026-10-07)): found peaks at the channel maxima on the correct sides (Co-60 400 µCi scene)

Req	Level	Method	Evidence	Status
SR-IMG-05	integration	T	`ImagingServiceTests.Strip_CoLocatedCsAndDoubleActivityCo_RecoversSameLiveTimeCsCount` (net count), `Strip_UsesSignedCcInput_AndMatchesMlemCountAndUncertainty`, `Strip_OverlapCovariance_ReplaysWithWindowChangesAndIncrementalPrefixes`, `Strip_MultipleContaminants_ReportNetWithUnavailableUncertainty`, `WindowChange_ReplaysRetainedEvents_RecalibratesAndMatchesFreshProcessing`; `StripCountEstimatorTests`, `StripProjectionTests`	pass (headless + offscreen); desktop pass 2026-10-07 (History (GCAM-SW-STR, Desktop verification — strip count, 2026-10-07))
SR-IMG-06	unit + integration	T, I	`ImagingWorkspaceTests.LateResponse_CannotReplaceNewerWindowResult`; serialized `Task.Run` worker and separate stopwatches	pass (headless); desktop pending
SR-IMG-07	unit + integration + offscreen	T, I	`ImagingWorkspaceTests.Reconstruction_IsAReprojectionSetting_KeepsMeasurementsAndSweep`, `ReconstructionNote_FlagsUnmeasuredOptics_AndTheAcquisitionImageIsNotShownAsMlem`, `StripCount_IsLabelledNetWithUncertainty_ForBothMethods`, `StripMetadata_AndRoiUnits_FollowPublishedViewWhileStripIsPending`; `ImagingServiceTests.Mlem_ChannelsLocalizeAtTheirOwnSource_NonNegative`, `Mlem_BuildsOneMatrixPerGeometryAndLine_AndReusesItAcrossRefreshes`, `CrossCorrelation_IsUnchangedByTheMethodField`; `MlemReconstructionTests` (factory MLEM bit-identical to the evidence construction); MLEM offscreen renders; imaging evidence	pass (headless + offscreen); desktop pass 2026-10-07 (History (GCAM-SW-STR, Desktop verification — strip count, 2026-10-07))
SR-RUN-01	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-09, -19
SR-RUN-02	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-10
SR-RUN-03	unit	T	`AcquisitionViewModelTests.Failure_WithData_KeepsItLocked_ResetOnly`, `Failure_WithoutData_BehavesAsEmpty`	pass
SR-RUN-04	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-11
SR-RUN-05	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-12
SR-RUN-06	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-13

Req	Level	Method	Evidence	Status
SR-RUN-07	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-14
SR-RUN-08	historical batch	—	batch implementation and batch-only tests removed; historical desktop evidence below	withdrawn → SR-RUN-15
SR-RUN-09	unit + integration	T, I	`AcquisitionViewModelTests.Reset_DiscardsData_UnlocksInputs_NextStartIsANewAcquisitionWithANewSeed`; `AcquisitionServiceTests.ShortAcquisition_LocalizesAfterCountThreshold_SnapshotsAreImmutable`; worker inspected	pass (headless); desktop pending
SR-RUN-10	unit + integration	T	`AcquisitionViewModelTests.Start_SnapshotsGrow_StopKeepsDataAndLocks_ContinueAccumulates`; `AcquisitionServiceTests.Stop_KeepsConsumedPrefix_AtExtremeMcLimitedSpeed`	pass (headless); desktop pending
SR-RUN-11	unit + integration	T	`AcquisitionViewModelTests.Completed_StartDisabledUntilPresetRaised_LowerPresetRejected_ProgressFollowsPreset`; `AcquisitionServiceTests.InjectedClock_AdvancesLiveTimeAndPresetWithoutWallDelay`	pass
SR-RUN-12	unit + offscreen + inspection	T, I	`AcquisitionViewModelTests.Start_CapturesDetectorInputs_AndLocksThemWhileDataExist` (every physical writer refused with data), `LiveTimeAndSpeed_LockedWhileAcquiring_InvalidValuesFallBack`; `AmbientViewModelTests.AmbientPhysicalInputsLockAndDerivedBsrHasNoSetter`; `OptionsViewModelTests.Draft_ChangesOwnerOnlyOnApply`, `InvalidDraft_StaysOpenWithFieldMessage_AndWritesNothing`, `LockedOptions_AreCloseOnly_AndDoNotApply`, `ModalHost_ReceivesOneDraft_AndCancelDiscardsIt`, `Summary_ShowsOnlyNondefaults_AndAmbientResultWithData`, `Summary_ListsEveryChangedOption_WithoutFixedFactsOrTopBarInputs`; `OptionsRenderTests.OptionsTabs_MovedIdsAndLocks_UseTheDraft_AndCloseOnlyWhenLocked`, `OptionsDialog_HasNoDefaultButton_EnterInAFieldKeepsTheDraftOpen` (no default button; a routed Enter in a field leaves the dialog open with the draft); render assertions: source editor and left-tab fields disabled with data, status-bar Options summary; `CanEditInputs` bindings inspected	pass (T/I); desktop re-run after stage 2 pending (`ChromeScenarioTests` drives the Options summary)

Req	Level	Method	Evidence	Status
SR-RUN-13	unit + inspection	T, I	`AcquisitionViewModelTests.Start_SnapshotsGrow_StopKeepsDataAndLocks_ContinueAccumulates` (Start / Continue label and availability), `Completed_StartDisabledUntilPresetRaised_LowerPresetRejected_ProgressFollowsPreset` (guard inside the command); `AmbientViewModelTests.DefaultIsValidatedFrontOnlyField_ZeroIsIdeal_SourceFreeStartRequiresAbsoluteField` (source-free Start with a field, disabled at 0); `StartRefusalTests` (eight refusal paths: optics, gap, scene, focus, readout not prepared, ambient / BSR with the physical readout, seed, ambient dose — status text, selected tab or focal section, cog warning until corrected, refusal cleared by a passing Start); `IsDefault` removed (I)	pass (T/I)
SR-RUN-14	removed	—	stale flag, chips and their tests removed	withdrawn → SR-RUN-12, SR-RUN-23, SR-RUN-26
SR-RUN-15	integration	T	existing nearest-prime / finite-grid builder tests; `AcquisitionServiceTests.InvalidInputs_FailBeforeStarting` (4 cases)	pass
SR-RUN-16	engine + integration	T, I	`ListModeSourceTests` (weighted flood, spectrum, rate, exponential gaps, seed, no duplicates); `CascadeEmissionTests` (biased Co-60 decays vs analog 4π decays: detected and coincident probability per decay and the rate within 4σ; one summed event per decay with one arrival time; a single-line config keeps its line); `CascadeSummingTests` (Co-60 branching, W(θ) moments and histogram, inverse CDF, Na-22 table values); service immutable / count conservation test	pass
SR-RUN-17	integration + inspection	T, I	injected-clock test (250 ms); acquisition ViewModel test (ROI grows, old snapshot unchanged); fixed grid and decode per tick inspected; timings in acquisition record	pass (T/I); desktop pending
SR-RUN-18	integration + unit	T	extreme MC-limited service test; virtual ViewModel test verifies note and retained prefix	pass
SR-RUN-19	unit + integration	T	Completed ViewModel test; service preset completion test; defaults inspected	pass

Req	Level	Method	Evidence	Status
SR-RUN-20	integration + unit + inspection	T, I	`DetectorRealismTests.StudioDefaults_AreExplicit_AndSceneBuilderRemainsBare`; `MainViewModelTests.DetectorInputs_DefaultsAndValidationFallbacks_EditableWithoutData`; `OptionsViewModelTests.Defaults_AreTheInitialValues_AndFallbacks_AndSummary`, `InvalidDraft_StaysOpenWithFieldMessage_AndWritesNothing`, `Summary_ListsEveryChangedOption_WithoutFixedFactsOrTopBarInputs`, `PhysicalMode_DisablesOnlyLegacyFields_AndMarksLegacyDifferencesInactive`; Options Detector tab bindings inspected	pass headless; desktop re-run after stage 2 pending
SR-RUN-21	integration + unit	T	`DetectorRealismTests.Gain_WidensPhotopeakInQuadrature_AndReducesTightWindowAcceptance`, `Measurement_UsesFrozenInputs_AndReplaysAcrossSnapshotsAndPileUp`; `AcquisitionViewModelTests.Start_CapturesDetectorInputs_AndLocksThemWhileDataExist`	pass
SR-RUN-22	engine + inspection	T, I	`ListModeBackgroundTests` verifies rate, spatial profile, energy response, exact disabled stream, seed and cancellation; `ReadoutBsrTests` (paired legacy / physical schedules, owned sites and pixel, zero-source contribution, cancellation in low-acceptance retries) and `ReadoutFieldIdentityTests` (recording-off BSR hashes); evidence readout-field-v2; BSR field and worker inspected	pass headless; desktop BSR input pending
SR-RUN-23	integration + unit	T	`AcquisitionContinuationTests.StopAndContinue_ReproducesTheUninterruptedEventStream` (3 cases, real session, one with Co-60 summed events), `Completed_RaisedPreset_ContinuesExactly`, `Continue_IsRejectedWhileRunning`; `OpticsViewModelTests.Focus_AcquiringStoppedCompleted_KeepsEventsAndSpectrum_OpticsLockedWithData` (one acquisition id across Continue)	pass
SR-RUN-24	unit	T	`AcquisitionViewModelTests.Completed_StartDisabledUntilPresetRaised_LowerPresetRejected_ProgressFollowsPreset`, `LiveTimeAndSpeed_LockedWhileAcquiring_InvalidValuesFallBack`	pass

Req	Level	Method	Evidence	Status
SR-RUN-25	integration + unit	T	`AcquisitionContinuationTests.Seed_FixedReproduces_OtherSeedIsAnIndependentAcquisition`; `AcquisitionViewModelTests.Reset_DiscardsData_UnlocksInputs_NextStartIsANewAcquisitionWithANewSeed`; seed on the Options Acquisition tab: `StartRefusalTests.SeedError_ShowsOptionsReason_WarnsOnTheCog`, `OptionsViewModelTests.Summary_ShowsOnlyNondefaults_AndAmbientResultWithData`	pass
SR-RUN-26	unit + offscreen + desktop	T, I	`AcquisitionViewModelTests.Reset_DiscardsData_UnlocksInputs_NextStartIsANewAcquisitionWithANewSeed`; `DeclutterTests.Reset_ConfirmsBeforeDiscard_OrKeepsData`, `Reset_WithoutConfirmation_IsDisabledAndKeepsData` (fake confirmation); `DeclutterRenderTests` (modal layer: single tree, Cancel default, Escape cancels); `DetectorWorkspaceTests.Face_FollowsPendingBeforeStart_AcquiredAndLockedAfter_PendingAgainAfterReset`; `FocusSweepViewModelTests.Sweep_RejectsLateResult_WhenIdentityChanges` ("reset"); confirmation renders (I); desktop `ScenarioTests.StopContinueReset_KeepsAccumulatesAndDiscards` (cancel keeps, discard resets)	pass (headless + offscreen); desktop pass 2026-10-12 (History (GCAM-SW-STR, Desktop verification — declutter stage 1, 2026-10-12))
SR-RUN-27	unit	T	`AcquisitionViewModelTests.Failure_WithData_KeepsItLocked_ResetOnly`, `Failure_WithoutData_BehavesAsEmpty`	pass
SR-RUN-28	unit + offscreen	T, I	`AcquisitionViewModelTests.Start_SnapshotsGrow_StopKeepsDataAndLocks_ContinueAccumulates` (status rate = counts / live time); status bar and Detector panel agree in the renders (84 vs 83.6 cps)	pass
SR-RUN-29	integration + unit + offscreen	T, I	`AmbientViewModelTests.DefaultIsValidatedFrontOnlyField_ZeroIsIdeal_SourceFreeStartRequiresAbsoluteField`, `PresetListOffersOnlyTheValidatedSpectrum_ByPinnedReference`, `Start_DefaultPassesValidatedPresetReference_ZeroTakesTheLegacyPath`, `AmbientPhysicalInputsLockAndDerivedBsrHasNoSetter`, `InvalidAmbientDoseIsRefused_PreviousValueStays`, `DoseTextOutsideThePattern_IsRefused_PreviousValueStays_StartBlocked` (11 entries), `DoseTextMatchingThePattern_SetsTheDoseRate` (6 entries), `DoseText_FollowsProgrammaticValue_AndLocksWithData`; `OptionsViewModelTests.InvalidDraft_St`	pass headless; offscreen PNGs inspected from a run past AN-15; desktop pass 2026-10-07 (History (GCAM-SW-STR, Desktop verification — 2026-10-07)): the whole desktop suite (34 / 34) runs with the ambient default on; desktop re-run after stage 2 pending (the inputs moved into the Options dialog)

Req	Level	Method	Evidence	Status
			`aysOpenWithFieldMessage_AndWritesNothing` (refused dose entry keeps the dialog open), `Summary_ShowsOnlyNondefaults_AndAmbientResultWithData` (ambient BSR in the summary with data); `StartRefusalTests.AmbientDoseError_ShowsOptionsReason_WarnsOnTheCog_PassingInputsClearTheRefusal`; `AmbientPresetTests.Pin_EqualsRepositorySidecar_AndDeployedBytesAreTheRepositoryFile`, `BuildConfig_ResolvesTheValidatedSpectrum_OnTheFrozenCopyOnly`, `WrongPinOrMissingFile_IsRefusedAtStart`, `DefaultField_AcquiresWithTheValidatedSpectrum`; `AmbientAcquisitionTests.BackgroundOnly_StopContinueRetainsExactlyTheFixedTimeStream`, `EmptyZeroField_CompletesAndBuildConfigFreezesInputs`, `BackgroundOnly_WorkspacesShowCountsWithoutInventingIsotopeLines`; offscreen Options Environment tab in the default and refused states (`ambient-inputs-*`, `ambient-inputs-invalid-*`) from `PlotViewRenderTests.Shell_OptionsCaptionStatusLeftTabs_Dark_RenderWithoutWindow` / `_Light_` (opt-in; before the 2026-10-13 split, the single `Spectrum_BothThemesFullAndZoom_RenderWithoutWindow`) inspected	
SR-CHAIN-01	integration + unit	T	`WaveformServiceTests.Response_AgreesAcrossWaveformSpectrumAndImaging`; `WaveformServiceTests.MixedFieldCalibration_UsesAcquiredChainAndInvalidatesRatios` (Evidence); `OptionsViewModelTests.Draft_ChangesOwnerOnlyOnApply` (chain chosen in the dialog, applied only on OK)	pass headless; desktop re-run after stage 2 pending
SR-CHAIN-02	unit + offscreen	T	`WaveformWorkspaceTests.Chain_LockedWithData_ScopeUsesAcquiredChain`, `ActiveAcquisition_DisablesPhysicalChainChanges`; `OptionsViewModelTests.PhysicalMode_DisablesOnlyLegacyFields_AndMarksLegacyDifferencesInactive`, `LockedOptions_AreCloseOnly_AndDoNotApply`; `OptionsRenderTests.OptionsTabs_MovedIdsAndLocks_UseTheDraft_AndCloseOnlyWhenLocked`	pass headless + offscreen

Req	Level	Method	Evidence	Status
SR-DET-01	unit	T	`DetectorWorkspaceTests.Gap_InvalidEditPreventsStart`, `Gap_RevalidatesAfterPitchChange_AndConvertsUnits`, `Gap_ZeroAllowsPitchBelowOldDefaultGap`; `OptionsViewModelTests.InvalidDraft_StaysOpenWithFieldMessage_AndWritesNothing`; `StartRefusalTests.GapError_ShowsOptionsReason_SelectsOpticsTab_WarnsOnTheCogUntilFixed`	pass headless; desktop re-run after stage 2 pending (`WorkspaceScenarioTests.Detector_FaceBeforeStart_LockedUntilReset` reads the gap through the Options dialog)
SR-DET-02, -03	unit + offscreen	T, I	`DetectorWorkspaceTests.Face_FollowsPendingBeforeStart_AcquiredAndLockedAfter_PendingAgainAfterReset`, `PanelRows_CarryTheReadoutFacts_PendingThenAcquired`; `DetectorFaceTests`; Detector renders before / after acquisition in both themes and sizes (`Workspaces_WaveformDetectorMeasurements_*`): identity line and gain legend rows in the right panel, face and colour bar only in the centre	pass headless + offscreen; desktop re-run after stage 2 pending (`WorkspaceScenarioTests.Detector_FaceBeforeStart_LockedUntilReset` reads `Detector.Identity`)
SR-DET-05	unit + integration	T	`ReadoutViewModelTests` (any valid geometry, per-line plans, eviction); `ReadoutOfferRecordTests`; `ReadoutBankTests`; `ReadoutLegacyIdentityTests` (Baseline 661.7 keV byte identity, all 144 gains)	pass (headless + offscreen); desktop pass 2026-10-12 (History (GCAM-SW-STR, Desktop verification — readout presets, 2026-10-12))
SR-DET-06	offscreen + unit	T, I	`ReadoutRenderTests` (SiPM / circuit view); engine circuit `ChargeDivisionNetworkTests`; `ReadoutViewModelTests.ViewSwitches_FollowTheReadoutMode_EmptyStateLeadsToTheReadoutChoice` (view tabs in every mode, headings, empty state, Choose readout), `CalibrationAndDiagnosticRows_CarryTheTextFacts`; `DetectorWorkspaceTests.PanelRows_CarryTheReadoutFacts_PendingThenAcquired`; view tabs and empty state asserted in the left-tab readout renders (`Shell_OptionsCaptionStatusLeftTabs_*`)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08)); view tabs and empty state pending on the desktop
SR-DET-07	integration	T	`PhysicalReadoutTests` (Prepare, cancellation, failure blocks Start, cache identity); `ReadoutViewModelTests.PreparationProgressAndOutcome_AreTheStatusBarText_ThenAcquisitionTakesOver`, `ReadyBank_TableRowsCarryTheReadyStateNumbers_OnlyWhenReady`; one progress element in the status bar asserted in the main-window renders (`AssertStatusBar`)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08)); status-bar progress pending on the desktop

Req	Level	Method	Evidence	Status
SR-DET-08	integration + offscreen	T, I	`PhysicalReadoutTests` (density / raw / LUT denominators); `ReadoutRenderTests`	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))
SR-DET-09	offscreen	T	`ReadoutRenderTests` (36 detached renders: states, tabs, four lanes, both themes, both sizes)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))
SR-DET-10	unit + integration + offscreen + desktop	T	`ReadoutViewModelTests` (localisation help built from the record: scope, module F range, direct-bias explanation, F wording; Wide FOV against the shipped record); `DeclutterTests.RemovedCostPresentation_HasNoViewModelMembersOrProfilePolicy`; `ReadoutRenderTests` (Calibration tab with mean sensitivity map); `DeclutterRenderTests` (cost section absent, physical channel controls in the channel section); `ReadoutOfferRecordTests`; desktop `ReadoutScenarioTests.WideLocalisationHelp_ShowsRecordScopeAndDirectBias`	pass (headless + offscreen); desktop pass 2026-10-12 (History (GCAM-SW-STR, Desktop verification — declutter stage 1, 2026-10-12))
SR-DET-11	unit + integration + desktop	T	`ReadoutViewModelTests` (Custom inherits offer by module; localisation help per module, none for an unmatched module); `ReadoutBankTests` (local Custom calibration, held facts once, per-line projection, flood correction); desktop `ReadoutScenarioTests.CustomModules_InheritByModuleNotCost_UnknownCalibratesLocally`	pass (headless + offscreen); desktop pass 2026-10-12 (History (GCAM-SW-STR, Desktop verification — declutter stage 1, 2026-10-12))
SR-WAVE-09	integration + offscreen	T	held replay and sum checks in `PhysicalReadoutTests`; trigger / hold placement and shared x in `ReadoutRenderTests`; `ReadoutViewModelTests.ViewSwitches_FollowTheReadoutMode_EmptyStateLeadsToTheReadoutChoice` (Energy sum / Four outputs tabs, empty state, the Σ-lane note)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08)); view tabs pending on the desktop
SR-WAVE-10	integration	T	`PhysicalReadoutTests` (stored record shared with Spectrum / Imaging)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))
SR-WAVE-11	integration	T	`PhysicalReadoutTests` (inactive workspace, disabled replay studies)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))

Req	Level	Method	Evidence	Status
SR-RUN-30	integration	T	`PhysicalReadoutTests` (immutable measured prefix, no re-measurement)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))
SR-RUN-31	unit + integration	T	`ReadoutStreamTests`; physical continuation in `PhysicalReadoutTests` (record-identical Stop / Continue incl. unfinished hold)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))
SR-RUN-32	integration	T	`PhysicalReadoutTests` (worker, cancellation, 4 Hz cadence, no invented counts)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08))
SR-RUN-33	integration	T	`PhysicalReadoutTests` (assigned / unknown / input counts, explicit keV windows, no stripping); `ReadoutViewModelTests.WindowPresets_ApplyEachWidthWithBenchmarkRounding_MarkFollowsValues` (4 cases), `BenchmarkWindowCommand_IsTheFifteenPercentPreset`, `BenchmarkWindow_SetsExactDecimalEdges`; window presets and their labels in the left Detector tab renders (`Shell_OptionsCaptionStatusLeftTabs_*`)	pass (headless + offscreen); desktop pass 2026-10-08 (History (GCAM-SW-STR, Desktop verification — physical readout, 2026-10-08)); window presets not yet run on the desktop
SR-RUN-34	unit + integration	T	`ReadoutViewModelTests` (field-only / missing preparation refused, incidental lines never block, failure recorded per line); `StartRefusalTests` (field-only reason; source + ambient / BSR routed after preparation); `ReadoutFieldAcquisitionTests` (both bounds, BSR and BSR + ambient: Stop / Continue identity, every line); `ReadoutBankTests` (service refuses an incomplete bank)	pass (headless + offscreen); desktop pass 2026-10-12 (History (GCAM-SW-STR, Desktop verification — readout presets, 2026-10-12))
SR-SCENE-01	unit	T	`MainViewModelTests.SourceItem_ClampsValues_LabelFollowsEdits`, `MainViewModelTests.IsotopePicker_OffersIr192_AndKeepsCs137AsTheDefault`	pass
SR-SCENE-02	unit	T	`MainViewModelTests.Startup_HasOneSelectedSource`, `MainViewModelTests.AddRemove_SelectsNewSourceThenNeighbour`	pass
SR-VIEW-01	unit	T	`HeatmapViewportTests.Fit_PreservesAspectAndCentres`	pass
SR-VIEW-02	unit	T, I	`HeatmapViewportTests.Snapping_FitsWholeDevicePixelsPerCell` (100 %, 125 %, sub-pixel); `HeatmapView` passes `PixelsPerDip` and `snapToWholePixels: true` (I)	pass

Req	Level	Method	Evidence	Status
SR-VIEW-03	unit	T	`HeatmapViewportTests.ZoomAt_KeepsAnchorPointFixed`, `HeatmapViewportTests.ZoomAt_IsClamped_FullZoomOutRefits`	pass
SR-VIEW-04	unit	T	`HeatmapViewportTests.PanBy_CannotDragImageOutOfView`	pass
SR-VIEW-05	unit + integration	T, I	`HeatmapViewportTests.ScreenImage_RoundTripWithYUp`, `HeatmapViewportTests.PixelAt_RespectsBoundsAndOrientation`, `HeatmapViewportTests.MmMapping_PutsPixelCentresOnGrid`; flood origin `-(N−1)/2·pitch` in `AcquisitionSession` (I), pinned by `AcquisitionServiceTests.FloodAxis_MatchesTheDecodersPixelCentres`	pass
SR-VIEW-06	unit	T	`HeatmapViewportTests.Configure_ResizeKeepsZoom_NewImageSizeRefits`	pass
SR-VIEW-07	system + unit	T, I, M	readout format and key handling in `HeatmapView` (I); value format `TickFormatterTests.Significant_FourDigitsGroupedWithoutTrailingZeros` (7 cases); readout exposed via `ItemStatus`; UI scenario `ScenarioTests.Readout_AndOneCellRoi_MatchAbsolutePositionAndValue` (§5)	partial — readout automated; the refit keys (double-click, `0`, Home) by inspection only
SR-VIEW-08	unit	T	`MainViewModelTests.Workspace_SharedResultAndSelectionSurviveSnapshot_ViewSettingsKeepState`; `ImagingWorkspaceTests.PeakChip_CountsSeveralFoundPeaks_AndNamesASingleOne`	pass
SR-VIEW-09	offscreen	I	colour bar and readout at the drawn image width (flood, reconstruction, Detector face) inspected in the 1280×800 / 1440×900 renders	pass offscreen; desktop deferred
SR-MEAS-01	unit + integration	T, M	`MeasurementMathTests.Distance_IsEuclidean`, `MeasurementsViewModelTests.Add_NumbersSequentially_AndSelectsTheNewOne`; VAL-03 (12.3 mm)	pass
SR-MEAS-02	unit	T, M	`MeasurementMathTests.AngleDeg_MeasuresAtTheVertex`, `MeasurementMathTests.AngleDeg_DegenerateArm_IsZero`; VAL-03 (67.0°)	pass
SR-MEAS-03	unit	T, M	`MeasurementMathTests.Roi_CountsPixelsByCentre_CornersInAnyOrder`, `MeasurementMathTests.Roi_ClipsToTheImage_AndIsEmptyBetweenCentres`; VAL-03 (ROI on flood)	pass

Req	Level	Method	Evidence	Status
SR-MEAS-04	unit + integration	T	`MeasurementsViewModelTests.Roi_HasNoValueBeforeARun_AndFollowsEachNewResult`, `MeasurementsViewModelTests.Roi_OnAPaneWithoutData_StaysEmpty`, `MeasurementsViewModelTests.Roi_WithNoPixelCentreInside_HasNoValue`, `MainViewModelTests.NewResult_RefreshesMeasurements`	pass
SR-MEAS-05	unit	T	`MeasurementsViewModelTests.Add_NumbersSequentially_AndSelectsTheNewOne`, `MeasurementsViewModelTests.Delete_SelectsTheNeighbour_ClearRestartsNumbering`, `MeasurementsViewModelTests.Add_WrongNumberOfPoints_Throws`, `Remove_DeletesThatRow_KeepsOrMovesSelectionLikeDelete` (per-row delete); row delete buttons and the trash can in the Measurements renders (`Workspaces_WaveformDetectorMeasurements_*`)	pass
SR-MEAS-06	unit	T	`MeasurementsViewModelTests.ToolHint_FollowsTheActiveTool`, `ActiveTool_IsOneOfFour_DefaultPan`; tool toggles inspected in the Measurements renders	pass
SR-MEAS-07	system	I, M	`MeasurementAdorner` (`MinDragPx = 4`, Esc / right-click / Delete handlers, `Clamp`); VAL-03 (Delete, Esc)	partial — 4 px threshold, right-click and clamping by I only
SR-MEAS-08	removed	—	Studio sets `CanMoveMarkers = False`; drag scenario rewritten	withdrawn → SR-RUN-12
SR-THEME-01	unit + offscreen + desktop	T, I	`MainViewModelTests.ThemeToggle_FlipsThemeAndRelabels` (with `FakeTheme`); `DeclutterRenderTests` (icon button tooltip and name); desktop `ScenarioTests.ThemeToggle_RelabelsAndSwitchesBack`	pass; desktop pass 2026-10-12 (History (GCAM-SW-STR, Desktop verification — declutter stage 1, 2026-10-12))
SR-THEME-02	offscreen + system	T, I, M	`ThemeService.Apply` replaces in place (I); `CaptionRenderTests.CaptionGlyphs_AreUnclipped_AndCaptionStatesUseThemeTokens`, `TopBarLine_RunsUnderTheCaptionButtons_WhichRestOnTheBarSurface`, `MaximizedWorkArea_UsesMonitorRelativePixels_AtEveryDpi` (3 DPI cases); top-bar order and chrome hit-testing asserted in the shell renders (`Shell_OptionsCaptionStatusLeftTabs_*`); desktop `ChromeScenarioTests.BorderlessCaption_HasNoNativeTitleBar_AndStartsAtWindowTop` (no UIA TitleBar element, top bar at the window top, maximise to the work area and restore) and the pilot's caption-button bounds — written, not yet run	partial — the both-theme visual swap has no dated manual record; the no-title-bar desktop assertion waits for the stage-2 desktop run

Req	Level	Method	Evidence	Status
SR-ENV-01	build	C, I	`Gcam.Studio.csproj` targets `net9.0-windows`, the other two `net9.0` (C); DWM return values ignored in `ThemeService.ApplyTitleBar`, which since 2026-10-13 affects only the remaining native frame (I)	pass — run on Windows 11 only
SR-ENV-02	system	I, M	`App.OnStartup` maximises when `SystemParameters.WorkArea` is smaller than the window (I); VAL-08	partial — VAL-08 open
SR-A11Y-01	system	I, M	30 `AutomationProperties.Name` in `MainWindow.xaml`, row name bound to `Description` (I); controls located by name in the VAL-03 session	pass
SR-A11Y-02	system	I, M	`HeatmapViewAutomationPeer` (I); heatmaps located by name + class `HeatmapView` in the VAL-03 session	pass
SR-A11Y-03	system	T, I, M	`HeatmapView.OnKeyDown` (I); tool access keys: `MainViewModelTests.MeasureToolAccessKeys_SelectTheTool_OnlyInImagingWithoutOptions`, `DeclutterRenderTests.ToolAccessKeys_AreWindowKeyBindings_SelectTheTool_OnlyInImagingWithoutModal` (the window's Alt+P / D / A / R bindings select the tool; inactive outside Imaging and behind a modal layer), tooltip and `AutomationProperties.AccessKey` declarations (`DeclutterRenderTests`, I), the `IconToggle` peer reporting the key (plain fact) and the rendered toggles' peers in the opt-in measurement renders; Enter: Start is not the default button (SR-RUN-13, I) and `OptionsRenderTests.OptionsDialog_HasNoDefaultButton_EnterInAFieldKeepsTheDraftOpen`	partial — real key routing and the keyboard-only walkthrough are desktop checks, not run (M); AN-01
SR-A11Y-04	system	I	`LiveSetting="Polite"` on the status line; selected row = accent bar + fill; selected overlay = thicker line + inverted chip	pass (I); no screen-reader check
SR-A11Y-05	unit + offscreen	T, I	`ThemeContrastTests` (8 cases against production tokens); acquired-seed foreground/fill assertions; both themes' 1280×800 renders inspected	pass: Surface 5.164:1 dark / 5.154:1 light; Canvas/Raised ≥4.5:1; desktop follow-up pending
SR-SEC-01	inspection	I	no HTTP or socket use in the three Studio projects; `SimulationService` passes an in-memory config (I, 2026-10-01); since TODO-46 the only `System.IO` use is `ReportFileWriter` and `ReportExportService`, which touch only the confirmed report and scene paths (I, 2026-10-15)	pass (I)

Req	Level	Method	Evidence	Status
SR-EXPORT-01	unit + offscreen + desktop	T, I	`ReportExportCommandTests.ExportReport_EnabledOnlyWithSnapshot_AndNotifies`, `ExportReport_CancelledDialogWritesNothing`; `ExportReportViewModelTests` (HTML default, event data always available, Export sends exactly the selected format, Cancel closes without exporting); dialog title, folder, names, filters and prompts in `ReportExportService` (I); offscreen title-bar and popup renders (`ExportSnapshots`); desktop `ReportExportScenarioTests`, one case per format	pass; desktop run 6 passed all four formats 2026-10-15 (History (GCAM-SW-STR))
SR-EXPORT-02	unit + desktop	T	`ReportWriterTests` (identical bytes and every section, escaping and no external reference, colormap, explicit missing views), `AcquisitionReportTests.Report_DefaultAmbientScene_HtmlOmitsSpectrumTables_SceneFileKeepsThem` (default scene: HTML 10.8 kB without figures, scene file keeps the full tables), `ReportExportCommandTests.ExportReport_CapturesBeforeDialogPumpsNewSnapshots`; `ReportDocumentTests.Pdf_HasPagesTextAndVectorFigures_WithoutImageObjects`, `Word_ContainsRealPngFigures_FromTheSharedSvgData`; `WordReportWriterTests.Write_ValidOoxmlPartsRelationshipsAndContent_AreDeterministic`; `ReportTextWrappingTests`; desktop: the HTML and PDF hold the displayed estimate, the Word document.xml holds it	pass
SR-EXPORT-03	unit + desktop	T	`ReportWriterTests.Write_SceneRoundTripsThroughConfigLoader_WithPhysicalReadoutNotice`, `AcquisitionReportTests`; desktop: the scene file loads	pass
SR-EXPORT-04	unit + desktop	T	`ReportFileWriterTests` (exact bytes, a failed sidecar removes a new report or keeps the existing pair), `ReportExportCommandTests.ExportReport_FailedWriteReportsFailure_WithoutChangingAcquisition`; desktop: exactly the declared files in the owned sandbox (two; three for event data)	pass

Req	Level	Method	Evidence	Status
SR-EXPORT-05	unit + desktop	T	`EventCsvWriterTests` (direct rows equal the retained events and measured energies in invariant culture; physical rows copy integer codes and assignments, refuse fractional codes; meta file names readout, columns, files and configuration; empty acquisition writes the header), `SpectrumServiceTests` (per-event measured energies re-histogram to the published spectrum bin for bin, with and without pile-up), `AcquisitionReportTests` (real direct acquisition: CSV rows equal the snapshot's events); desktop: the CSV case reads the header, six cells per row and the meta file	pass
SR-CMP-01 … 05	unit + offscreen + desktop	T, I	`ReadoutCompareServiceTests` (retained-input projections, populations from a hand-built fixture, disagreement denominators, line-specific overlay, coalescing and revision rejection), `CompareWorkspaceTests` (states, activation, export-omission note), `ReadoutLegacyIdentityTests` and `ReadoutDefaultPathTests` (existing DirectCrystal and readout hashes unchanged); offscreen `CompareSnapshots` (nine states and a real Baseline acquisition, both themes, 1280×800 / 1440×900); desktop `CompareScenarioTests.PhysicalPrefix_ComparisonDifferenceAndReset_KeepTheirPopulationLabels`	pass headless + offscreen + desktop (2026-10-15)
SR-ARCH-01	build	C	`Gcam.Studio.Core.csproj` targets plain `net9.0` without `UseWPF`: a WPF type does not compile	pass
SR-ARCH-02	build	C, I	Core has no reference to `Gcam.Simulation` (C); the shell could reach it transitively through Services, so for the shell it is I only	pass — see AN-08
SR-ARCH-03	inspection	I	`MainWindow.xaml.cs` is `InitializeComponent()` only; cited literal sizes replaced by theme metrics — AN-05 closed	pass by inspection
SR-ARCH-04	build	C, I	`Gcam.Studio.Tests.csproj` references only `Gcam.Studio.Core` and targets `net9.0`	pass
SR-NAV-01	unit + I	T, I	`MainViewModelTests.Workspace_SharedResultAndSelectionSurviveSnapshot_ViewSettingsKeepState`; workspace-type centre / panel templates	pass
SR-NAV-02	unit + system	T, I	workspace tests, `WorkspaceActivation_SelectsShell_WhenActiveIsSetWithoutCommand` and `Spectrum_SnapshotsGrow_ViewSettingsReuseAcquisition_SelectionSurvives`: checked-state activation, identities, unavailable index and retained selection; segmented switch / Ctrl+1…4 inspected	partial — desktop SelectionItem switching pending

Req	Level	Method	Evidence	Status
SR-NAV-03	unit	T	`AcquisitionViewModelTests.Start_CapturesDetectorInputs_AndLocksThemWhileDataExist` (physical writers refused, window N still applied), `Workspace_SharedResultAndSelectionSurviveSnapshot_ViewSettingsKeepState`	pass
SR-PLOT-01	unit + I	T, I	`PlotSeriesTests.LowerBound_HandlesUniformAndIrregularX_AndEnds`; validation / cap / band / marker properties inspected	pass
SR-PLOT-02	unit	T	`PlotViewportTests.Mapping_RoundTrips_AndClampsLogFloor`, `NiceTicksTests.Linear_Uses125Steps_EngineeringLabels_NoNegativeZero`, `Logarithmic_LabelsDecades_WithEightMinorTicks`, `TickFormatterTests.StepLabels_TakeDecimalsFromTheStep_AndGroupThousands`	pass
SR-PLOT-03	unit	T	`MinMaxPyramidTests.Query_EqualsBruteForce_IncludingBothEnds` (5 cases), `Range_ClipsOutsideData_RejectsNonFiniteSamples`	pass
SR-PLOT-04	unit + system	T, I	`PlotViewportTests.ZoomPanReset_PreservesAnchor_AndBounds`; `PlotViewTests.TenMillionSamples_ZoomAndResizeRedraw_Within16Milliseconds` checks production peer, + key and reset; pointer / other keys / readout by inspection	partial — full input walkthrough pending
SR-PLOT-05	system	T	`PlotViewTests.TenMillionSamples_ZoomAndResizeRedraw_Within16Milliseconds`; measured record in §5	pass (CPU redraw only)
SR-PLOT-06	unit + offscreen render	T	`HistogramPlotTests.Steps_IncludePartialBins_AndEmptyBinHasFullWidth`, `Envelope_RetainsExtremaOfBinsCrossingColumns`, histogram validation; `PlotViewRenderTests` PNGs	pass headless
SR-PLOT-07	unit + offscreen render	T	`Configure_PreservesZoomAndPanUntilRangeChanges_AndResetStillFits`; acquisition snapshot VM test; render test replaces counts under zoom	pass headless
SR-PLOT-08	unit + offscreen render	T	`VisibleExtent_ExcludesDistantPeak_IncludesIntersectingBins`; render test checks live top retention	pass headless
SR-PLOT-09	unit + inspection	T, I	`BinLookup_UsesHalfOpenEdges_AndReadoutUsesCountsAndUnits`; `PlotView.UpdateHover` and drawing inspected	pass headless; desktop pointer walkthrough deferred
SR-PLOT-10	unit + offscreen render	T, I	`Labels_ClampBothEdges_AndUseAdditionalRowsWithoutCollisions`; Spectrum, Waveform and focus-curve renders show band / marker labels in the strip above the data, the shaped waveform plot without marker labels	pass headless

Req	Level	Method	Evidence	Status
SR-PLOT-11	offscreen render	T	`PlotViewRenderTests.Spectrum_BothThemesFullAndZoom_RenderWithoutWindow`, independent opt-in, four PNGs	pass
SR-SPEC-01	integration + I	T, I	`SpectrumServiceTests.CsAcquisition_PhotopeakBinAndFwhmMatchChain`, `Axis_IsFixedAt2000keV_ForAnyLinesFieldOrPileUp_AndOverflowKeepsEveryPulse`; `MainViewModelTests.SpectrumSummary_NamesTheOverflowAndTheFixedAxisEnd`; Histogram binding / explicit edges, axis labels and no synthetic noise inspected	pass headless; retained four-workspace desktop survey; no pixel pass/fail oracle
SR-SPEC-02	integration	T	`CsAcquisition_PhotopeakBinAndFwhmMatchChain`; engine `FrontEndPartsTests` verifies legacy chain and pulse compatibility	pass
SR-SPEC-03	integration	T	`HighRate_PileUpLosesPulsesAndMovesCountsAbovePhotopeak`; exact pulse-count agreement with engine `ApplyPileUp`	pass
SR-SPEC-04	integration + unit	T	`MixedCsCo_HasFourBandsAndUnionCountsEachBinOnce`, `Merge_UsesResolutionRatherThanWindowOverlap` (2 cases), `Merge_SingleLineGivesOneBand`	pass
SR-SPEC-05	integration + unit + offscreen	T, I	`MixedCsCo_HasFourBandsAndUnionCountsEachBinOnce`; `SpectrumBandTests.BandOfXRayLines_IsNamedByEmitter_GammaBandByIsotope`; engine `EmissionKindTests` (3 cases); `VerifyEmissionTable` asserts shared header/row column edges, full strings and ≥8-DIP gaps, including both Co-60 windows and grouped counts at both sizes/themes	pass headless/offscreen; updated table desktop pending
SR-SPEC-06	unit + I	T, I	`AcquisitionViewModelTests.Spectrum_SnapshotsGrow_ViewSettingsReuseAcquisition_SelectionSurvives` (acquisition and its state unchanged)	pass headless and final desktop band/window scenario
SR-SPEC-07	integration + I	T, I	`SeedAndSnapshotPartition_AreDeterministic_ToggleReplaysExactly` (2 cases), `Processing_MeasuresFullAndIncrementalWorkAt100000Events`; service `Task.Run` inspected	pass
SR-SPEC-08	inspection + system	I	`SpectrumView.xaml`, `SpectrumPanel.xaml`, inherited `PlotView` peer	partial — both themes surveyed; screen-reader/keyboard-only walkthrough pending
SR-SPEC-09	unit + integration + inspection	T, I	acquisition VM test checks range request / selection persistence; spectrum service verifies every bin edge / centre; XAML one-way range / units inspected	pass headless; desktop selection walkthrough deferred

The generated table below reports the selected automated test executions only. The matrix's inspection, compiler and manual judgments and its desktop statuses stay separately reviewed evidence; passing automated cases alone do not sign off a requirement.

Automated execution only; inspection/compiler/manual verdicts in the matrix remain separate.

SR / EV / VAL	Passed	Failed	Not executed	Missing methods
EV-01	10	0	0	0
EV-02	4	0	0	0
EV-03	7	0	0	0
EV-04	5	0	0	0
EV-05	1	0	0	0
EV-07	18	0	0	0
EV-08	10	0	0	0
EV-10	2	0	0	0
EV-11	37	0	0	0
EV-12	7	0	0	0
EV-13	1	0	0	0
EV-14	4	0	0	0
EV-15	24	0	0	0
EV-17	15	0	0	0
EV-19	12	0	0	0
EV-20	20	0	0	0
EV-22	5	0	0	0
EV-23	6	0	0	0
EV-25	6	0	0	0
EV-29	6	0	0	0
EV-30	5	0	0	0
EV-31	4	0	0	0
EV-32	4	0	0	0
EV-33	10	0	0	0
EV-34	87	0	0	0
EV-35	16	0	0	0
SR-CHAIN-02	1	0	0	0
SR-CMP-01	1	0	0	0
SR-CMP-02	1	0	0	0
SR-CMP-03	1	0	0	0
SR-CMP-05	1	0	0	0
SR-DET-01	1	0	0	0
SR-DET-10	1	0	0	0

SR / EV / VAL	Passed	Failed	Not executed	Missing methods
SR-DET-11	1	0	0	0
SR-EXPORT-01	1	0	0	0
SR-EXPORT-02	1	0	0	0
SR-EXPORT-03	1	0	0	0
SR-EXPORT-05	1	0	0	0
SR-IMG-01	2	0	0	0
SR-IMG-02	11	0	1	0
SR-IMG-03	1	0	0	0
SR-IMG-04	7	0	1	0
SR-IMG-05	15	0	0	0
SR-IMG-06	1	0	0	0
SR-IMG-07	24	0	0	0
SR-MEAS-01	3	0	0	0
SR-MEAS-02	4	0	0	0
SR-MEAS-03	5	0	0	0
SR-MEAS-04	4	0	0	0
SR-MEAS-05	4	0	0	0
SR-MEAS-06	2	0	0	0
SR-MEAS-07	1	0	0	0
SR-NAV-01	1	0	0	0
SR-NAV-02	3	0	0	0
SR-NAV-03	2	0	0	0
SR-OPT-01	2	0	0	0
SR-OPT-02	3	0	0	0
SR-OPT-03	17	0	0	0
SR-OPT-04	7	0	0	0
SR-OPT-05	3	0	0	0
SR-OPT-06	0	0	4	0
SR-PLOT-01	1	0	0	0
SR-PLOT-02	5	0	0	0
SR-PLOT-03	10	0	0	0
SR-PLOT-04	2	0	0	0
SR-PLOT-05	1	0	0	0
SR-PLOT-06	2	0	1	0

SR / EV / VAL	Passed	Failed	Not executed	Missing methods
SR-PLOT-07	1	0	0	0
SR-PLOT-08	1	0	0	0
SR-PLOT-09	1	0	0	0
SR-PLOT-10	1	0	0	0
SR-PLOT-11	0	0	1	0
SR-RUN-03	2	0	0	0
SR-RUN-09	3	0	0	0
SR-RUN-10	3	0	0	0
SR-RUN-11	2	0	0	0
SR-RUN-12	17	0	0	0
SR-RUN-13	16	0	0	0
SR-RUN-15	4	0	0	0
SR-RUN-16	20	0	0	0
SR-RUN-20	10	0	0	0
SR-RUN-21	3	0	0	0
SR-RUN-22	6	0	0	0
SR-RUN-23	8	0	0	0
SR-RUN-24	2	0	0	0
SR-RUN-25	2	0	0	0
SR-RUN-26	12	0	0	0
SR-RUN-27	2	0	0	0
SR-RUN-28	1	0	0	0
SR-RUN-29	41	0	2	0
SR-RUN-31	9	0	0	0
SR-RUN-34	4	0	0	0
SR-SCENE-01	2	0	0	0
SR-SCENE-02	2	0	0	0
SR-SPEC-01	3	0	0	0
SR-SPEC-02	3	0	0	0
SR-SPEC-03	1	0	0	0
SR-SPEC-04	5	0	0	0
SR-SPEC-05	5	0	0	0
SR-SPEC-06	1	0	0	0
SR-SPEC-07	3	0	0	0

SR / EV / VAL	Passed	Failed	Not executed	Missing methods
SR-THEME-01	2	0	0	0
SR-THEME-02	4	0	0	0
SR-VIEW-01	1	0	0	0
SR-VIEW-02	3	0	0	0
SR-VIEW-03	2	0	0	0
SR-VIEW-04	1	0	0	0
SR-VIEW-05	5	0	0	0
SR-VIEW-06	1	0	0	0
SR-VIEW-07	8	0	0	0
SR-VIEW-08	3	0	0	0
VAL-01	1	0	0	0
VAL-02	1	0	0	0
VAL-03	9	0	0	0
VAL-04	1	0	0	0
VAL-05	1	0	0	0

Coverage summary

The tables below preserve the pre-acquisition baseline (47 rows); withdrawn RUN rows are historical evidence. The acquisition, spectrum and additional plot rows are verified above by tests / inspection and retained offscreen rendering. The final four-workspace desktop pass below adds current acquisition and workspace-path evidence; keyboard-only, screen-reader and SelectionItem coverage remain incomplete; the prior batch desktop passes must not be treated as acquisition validation.

Primary method	Requirements
Automated test (alone or with I / M)	33 — NAV 01–03, PLOT 01–05, RUN 01–08, SCENE 01–02, VIEW 01–06 + 08, MEAS 01–06 + 08, THEME-01
Manual UI Automation + inspection	7 — VIEW-07, MEAS-07, THEME-02, ENV-02, A11Y-01, A11Y-02, A11Y-03
Inspection only	3 — A11Y-04, SEC-01, ARCH-03
Compiler-enforced (+ inspection)	4 — ENV-01, ARCH-01, ARCH-02, ARCH-04

Status	Count
pass	38
partial	9 — NAV-02, PLOT-04, RUN-04, VIEW-07, MEAS-07, THEME-02, ENV-02, A11Y-03, ARCH-03
open	0 requirements; system-level scenarios are open in §5

Spectrum physics and processing evidence

`SpectrumServiceTests` acquires fresh real-engine events with transport seed 12345 and processing seed 909. The default chain is GAGG(Ce) / Hamamatsu MPPC S13360-3050 / CSP + CR-RC, with 4.5692%

FWHM at 662 keV.

Seeded values re-measured 2026-10-02 after the random generator was replaced (old generator: 31,053 deposits; Ba ratios 0.480119 / 0.626183; FWHM 30.3436 keV; pile-up 85,108 / 4,735 / 456; union share 37.1900 %); every check passes with its unchanged tolerance. The bin-dependent values (photopeak bin, FWHM, pile-up counts, union share) are re-measured on the fixed 0–2000 keV axis of 2 keV bins (2026-10-04); the physics checks are unchanged.

Check	Measurement	Acceptance / tolerance
Cs-137 photopeak, 100,000 events (31,300 full-energy deposits)	regional maximum bin 330, containing 661.7 keV; bin width 2.0000 keV	exact bin match, maximum above 478 keV through axis end
Ba K absorber response	uncollided ratio 0.474932 vs narrow-beam 0.474899; measured-band ratio 0.619148 vs independent weighted MC 0.619379	4σ tolerances 0.014987 and 0.028971; global peak ordering is reported rather than asserted; detailed conditions in acquisition evidence
Cs photopeak FWHM	histogram 30.2798 keV vs `FrontEndModel` 30.2373 keV; error 0.0425 keV	≤4.6043 keV = two bins + 5·FWHM/√(2(N−1)), N = full-energy deposits
High-rate Cs pile-up (activity 10^6 µCi, 100,000 events)	100,000 → 85,217 pulses; above 794.04 keV: 0 → 5,265; overflow 50; resolving time 730 ns	strictly fewer pulses and more counts above 1.2×661.7; pulse count exactly matches `ApplyPileUp` at 1 ps time rounding
Cs + Co, 20,000 events	four bands: 32.1 + 36.4, 661.7, 1173.2, 1332.5 keV; union share 37.0650%	exact labels / band count and union share; default Co windows need not overlap
Merge unit fixtures	Ba K merged; single line retained; injected constant 10% FWHM keeps 1173.2 / 1332.5 separate despite overlapping N=1.5 windows	exact band counts; separation 159.3 keV > FWHM(mean) 125.285 keV; fixture is not a physical-chain claim
Determinism, 10,000 events	batch, prefix 503 + remainder, and toggle replay agree with / without pile-up	exact histogram and overflow equality
`Axis_IsFixedAt2000keV_ForAnyLinesFieldOrPileUp_AndOverflowKeepsEveryPulse`	1000 bins of 2 keV over 0–2000 keV for 3 line lists (Cs, Co, both) × 2 field maxima (none, 3960.9 keV) × pile-up off / on; 2505.7 and 3000 keV pulses in the overflow; total = drawn + overflow; histogram and 662 band identical with and without the field maximum	exact

Worker elapsed time measured by `Stopwatch` inside `Task.Run` (five warmed Release runs; excludes queue / gate waiting, includes worker preemption). No hard timing tolerance is imposed on this measurement.

Work	Five measurements (ms)
Smear + bin full 100,000-event snapshot	102.166, 176.062, 101.702, 116.910, 101.117
Increment 1,000 events over cached 99,000	1.267, 1.054, 1.034, 1.323, 1.065

Work	Five measurements (ms)
Pile-up toggle, replay 100,000 events	135.560, 102.186, 102.762, 112.295, 101.615

Full processing exceeds a few ms, so it stays on the service worker. This is processing evidence, not a desktop responsiveness measurement. Once the desktop and UI-test project are free, add live-count growth / Stop retention, Cs photopeak / Ba K and mixed four-band display, setting reuse, workspace switching and selection retention, plot keyboard / readout / focus, accessible control and table names, and both-theme checks.

5. Validation

User-level scenarios run on the real app. "Performed" means driven through UI Automation with real pointer and key input (`SetCursorPos` + `mouse_event`): controls are found by `AutomationId`, the run state is read from the status line's `ItemStatus`, and each verdict is judged on the app's own state against a value computed independently of the app's code (an oracle that re-derives the image layout and the screen → mm mapping from the documented rules).

ID	Scenario	Acceptance criteria	Status
VAL-01	Start the default scene and read the result	Completed status, flood and reconstruction present, counts / live time shown	Performed 2026-10-02: pilot passes
VAL-02	Stop, Continue and Reset an acquisition	Stop freezes counts and retains locked data; Continue adds counts under one acquisition; Reset discards and unlocks	Performed 2026-10-02: rewritten acquisition scenario passes
VAL-03	Measure on both images	distance and ROI drawn on the flood map, angle on the reconstruction; values appear in "Measurement results" and on the image; Delete removes the selected one; Esc abandons a half-drawn angle	**performed 2026-10-01** — by hand: distance 12.3 mm and an ROI on the flood map, angle 67.0° on the reconstruction, Delete and Esc as specified; automated: three scenarios below
VAL-04	Move a source after an acquisition	Data lock fields / markers; Reset unlocks; edit X / Y and Start; found peak follows the new position	Performed 2026-10-02: reset/edit/start scenario passes
VAL-05	Locate a source and measure its offset from the decoded peak	Distance tool from the source marker to the peak gives the offset in mm; it agrees with the peak chip and source X / Y to within one recon step	**partial 2026-10-01** — automated: after moving the source and re-acquiring, the decoded peak lay within 1.5 mm of the source; the Distance-tool step is not automated
VAL-06	Use both themes	Text, chips, plots and overlays inspected in all four workspaces at two sizes	Performed 2026-10-02: 16-frame desktop survey; faint locked inputs recorded; keyboard-focus walkthrough excluded
VAL-07	Keyboard only	every function reachable without a pointer except creating measurements (AN-01)	open
VAL-08	Small screen and display scaling	at 1366×768 the window starts maximised; at 125 % / 150 % heatmap cells stay equal width	open

Plot performance gate

2026-10-01, local verification host, Windows 11 build 26200, .NET SDK 9.0.311 / runtime 9.0.13, Release, 100% DPI, 10,000,000 samples, visible 1280×800 host resized to 1440 width. The tool shell reports 24 processors; `DOTNET_PROCESSOR_COUNT=4` constrained build / test runtime concurrency. Three warm-up zoom redraws preceded five measured zoom / resize pairs. Data preparation is outside the interactive redraw measurement.

CPU redraw	Five measurements (ms)	Maximum	Gate
Zoom	2.0368, 12.1177, 3.2046, 1.0483, 1.3064	12.118 ms	≤16 ms: pass
Resize	2.6021, 2.6420, 0.7743, 0.7808, 0.6819	2.642 ms	≤16 ms: pass

CPU `OnRender` includes axes / ticks, exact extrema query, frozen geometry and WPF drawing commands. It excludes dispatcher scheduling and compositor presentation. Event-to-render resize latency was 280.517… 315.842 ms; this is **not** evidence of a ≤16 ms end-to-end resize. Full record: performance.json. No data were removed or extrema dropped. The host test also checks the production automation peer and keyboard + / reset.

Latest desktop regression and polish survey attempt

The opt-in Release run on 2026-10-01 returned **11 pass / 8 fail / 0 skipped**: 10 oracle cases and the plot gate passed; all seven existing scenarios and the survey failed. The pilot reached Succeeded and found the flood map, then `SetCursorPos` failed. The remaining scenarios failed at UIA `SetFocus` while normalising the owned window. Screen-copy diagnostics also failed (invalid handle). Existing assertions were not weakened. All owned app processes exited with 0 and their sandbox-write lists were empty. This does not establish regression success.

Failure manifests / tree dumps are under the ignored test output `ui-runs/20261001-154607-7bf65f/` through `ui-runs/20261001-154654-dfd8b1/`. The dated earlier scenario results below are historical evidence. The four requested Imaging captures and observed polish issue list are **pending**, as is a keyboard walkthrough; survey record and intended paths. No screenshot or issue is invented.

Automated scenarios (`tests/Gcam.Studio.UiTests`, opt-in with `GCAM_UI_TESTS=1`)

They move the real mouse, so they run only on a free interactive desktop; in an ordinary `dotnet test` and in CI they are skipped and reported as skipped, not passed. Each scenario starts its own sandboxed app.

Scenario	Traces to	Oracle (independent of the app)	Result 2026-10-01
`PilotTests.Pilot_SimulateThenMeasureDistance_AddsRowWithExpectedLength`	VAL-01, VAL-03 · SR-RUN-09, SR-MEAS-01	length from screen points through the documented fit rule	13.4 mm vs 13.400 ± 0.204

Scenario	Traces to	Oracle (independent of the app)	Result 2026-10-01
`ScenarioTests.StopContinueReset_KeepsAccumulatesAndDiscards`	VAL-02 · SR-RUN-10, SR-RUN-12, SR-RUN-13, SR-RUN-23, SR-RUN-26	state machine Acquiring → Stopped → Acquiring (Continue) → Stopped → Reset → Empty; counts frozen after Stop, growing after Continue, zero after Reset; inputs locked until Reset	pass 2026-10-02: frozen after Stop, accumulated after Continue, unlocked after Reset
`ScenarioTests.Roi_OnFloodMap_CountsWholePixelsByCentre`	VAL-03 · SR-MEAS-03	pixel count by centre, corners on cell boundaries so one pixel of error cannot change it	42 px, 3.6 × 4.2 mm
`ScenarioTests.Angle_EscAbandonsDraft_DeleteRemovesSelected`	VAL-03 · SR-MEAS-02, SR-MEAS-07	angle from the three screen points; Esc is proven by the value	69.8° vs 70.02 ± 2.29; Delete removed it
`ScenarioTests.MoveSource_ResetEditStart_PutsPeakOnTheSource`	VAL-04, VAL-05 · SR-RUN-12, SR-RUN-26, SR-VIEW-08	physics: after Reset, an X / Y edit and a new acquisition the decoded peak must sit on the moved source (≤ 1.5 mm)	pass 2026-10-02: peak (21.6, 17.1), source (21.4, 16.0) mm; existing 1.5 mm tolerance
`ScenarioTests.Readout_AndOneCellRoi_MatchAbsolutePositionAndValue`	SR-VIEW-05, SR-VIEW-07, SR-MEAS-03	absolute mm of two cells (both signs); a one-cell ROI must sum to the readout's value	(−6.3, 5.1) and (6.3, −4.5) mm exact; sum = cell value
`ScenarioTests.ThemeToggle_RelabelsAndSwitchesBack`	SR-THEME-01	the label names the other theme; the app still simulates after two swaps	pass (the visual swap stays manual)
`ChromeScenarioTests.BorderlessCaption_HasNoNativeTitleBar_AndStartsAtWindowTop`	SR-THEME-02, SR-RUN-12	no UIA `TitleBar` element in the window; top bar and Close within 1 px of the window's top and right edges; the Options summary follows an applied seed and ignores a cancelled one; Maximize fills the monitor work area (Win32 monitor info), Restore returns	desktop pass 2026-10-13 (History (GCAM-SW-STR, Desktop verification — declutter stage 2 and TODO-42, 2026-10-13))

Record (list-mode acquisition, 2026-10-01): all 19 UI tests passed (10 oracle, 7 desktop scenarios, the plot gate, the polish survey); with every scenario's expectation deliberately corrupted (`GCAM_UI_BREAK_VERDICT=1`) all 7 desktop scenarios failed, each at its corrupted assertion, so the verdicts can fail. No app process or sandbox folder was left behind. UIA Select on the workspace switch does not change the workspace (see anomalies).

6. Known anomalies, gaps and risks

Anomalies and gaps

ID	Finding	Effect	Plan
AN-01	Measurements can only be **created** with the pointer (documented in DESIGN.Controls)	keyboard users can review, select and delete, not create	keyboard crosshair
AN-02	UI automation is desktop-only and opt-in	the 26 regression scenarios (29 desktop cases with the pilot, plot gate and survey) are not part of CI; keyboard-only flows and other display scaling remain unverified	heatmap navigation scenarios next

ID	Finding	Effect	Plan
AN-03	No high-contrast mode	Windows high-contrast themes are not honoured	planned with the keyboard crosshair
AN-04	Closed: legacy batch progress race	unused batch progress implementation removed; acquisition uses sequential snapshots	batch regression removed with the code it guarded; acquisition progress verified by `AcquisitionViewModelTests.Completed_StartDisabledUntilPresetRaised_LowerPresetRejected_ProgressFollowsPreset`
AN-05	Closed: the cited literal sizes in `MainWindow.xaml` (photon box `Width="120"`, readout `Height="18"`, `Margin="0,4,0,0"`, colour-bar `Margin="0,8,0,0"`) were replaced by theme metrics	original spacing defect retained here as history	current MainWindow uses named metric resources
AN-06	Closed (2026-10-02): the "outdated" chip was removed with the stale state	physical inputs are locked while data exist, so images always match the inputs shown; the state is announced by the status line (live region)	—
AN-07	WPF items (`HeatmapView` rendering, `ColorBar` ticks, `ThemeService`) have no unit tests, by design	defects show only when the app runs	covered by AN-02 plan
AN-08	SDK-style project references are transitive, so the shell *could* call the engine directly	layering rule for the shell is inspection-only	optional: `PrivateAssets` on the Services → engine reference
AN-09	*Closed* — the photon budget no longer exists (list-mode acquisition)	—	—
AN-10	Workspace checked-state activation now selects the shell workspace; `WorkspaceActivation_SelectsShell_WhenActiveIsSetWithoutCommand` passes	screen-reader SelectionItem switching still needs desktop verification	ViewModel fix implemented; add a desktop Select regression and verify the displayed workspace, without relying on the checked state alone
AN-11	The source-drag scenario localised at 1.43 mm against its 1.5 mm tolerance. **Cause found (2026-10-02):** with the default optics at a 1 m focal plane the detector samples the mask-cell shadow at 1.27 pixels per cell (0.761 mm shadow, 0.6 mm pixels), below Nyquist, so the decoded peak shifts with the source position (RMS 0.95 mm, max 1.91 mm over y = −10 … 10 mm; 0.24 / 0.41 mm at 3.8 samples per cell) — see VV.Studio.Imaging	source drag is withdrawn; the current scenario verifies Reset / field edit / new acquisition, while undersampling remains a physical limitation	use optics with ≥ 2 samples per cell at the working focal plane; retain the localisation tolerance
AN-12	Closed: desktop input and screen capture available in the final pass (2026-10-02)	all scenarios, the plot gate and four-workspace survey executed	evidence below

ID	Finding	Effect	Plan
AN-13	CPU plot redraw meets the gate, but resize event-to-render delay is materially larger	CPU evidence does not establish end-to-end responsiveness	distinguish timings; investigate dispatcher / desktop latency before making a presentation-latency claim
AN-14	`TickFormatterTests` (colour-bar tick labels: one shared multiplier, one decimal count, no negative zero) test behaviour that no SRS row states	the behaviour is verified but not required, so a change to it would not be traced	add an `SR-VIEW` row for colour-bar labels
AN-15	The opt-in offscreen render test (then the single `PlotViewRenderTests.Spectrum_BothThemesFullAndZoom_RenderWithoutWindow`; since the 2026-10-13 split the emission-table check runs in `MainWindow_Workspaces_*`, `MainWindow_MixedField_*` and `MainWindow_FocalPlaneAndMlem_*`) fails on one machine (2026-10-04) in its emission-table check ("Truncated table value: Window (keV)") before it reaches the later renders; it fails the same way at a commit before the ambient default and fixed axis, so it predates them (likely a font or display difference on that machine — not established)	on that machine the repository test does not complete SR-SPEC-05's table check or produce the renders after it; the ambient-panel PNGs (SR-RUN-29) came from a scratch copy that logged that assertion instead	find why the window column truncates there, then re-run the opt-in render test unchanged
AN-16	Closed (2026-10-13): SR-A11Y-03 named tool access keys (Alt+P / D / A / R) and "Enter runs (default button)" after the access keys were lost with the icon toggles and Start had stopped being the default button (SR-RUN-13)	the access keys are restored as window key bindings (tooltips name them, the toggles expose them as their accessible access key) and SR-A11Y-03 states the shipped Enter behaviour	key routing on the real window is verified with the next desktop run
AN-17	Closed (2026-10-13): the desktop run found that the modal focus boundary walked only the visual tree and refused the focus an opening combo gives its drop-down item (a popup root has no visual parent), so the Options dialog's combos (ambient bound, detection chain) could not be chosen through their drop-down and a UI scenario failed ("No FrontOnlyThroughMask in Acquisition.AmbientBound")	Options inputs set through a combo were unusable	`ModalHost` follows the logical parent and popup placement target; `ModalFocusRenderTests` checks it windowlessly (fails on the old boundary); `Choose` requires an open drop-down; desktop re-run passes

ID	Finding	Effect	Plan
AN-18	`CompareWorkspaceTests.SettingsAndReset_RejectOldResults_EvenWhenServiceIgnoresCancellation` failed once in milestone M3 (2026-10-17, its 5 s wait on `WhenUpdated` timed out) and passed in 20 / 20 repeats of the whole assembly. Likely cause, from the code: setting `WindowLowKeV` raises two property changes, so `Refresh` runs twice; the tests have no synchronisation context, so `UpdateAsync` continues on the thread pool and can take `_pending` before it records `_running`, letting the second `Refresh` queue the same key again — a second request the fixture never completes	a rare red result in a test run; in Studio both run on the UI thread, so the duplicate cannot occur there	run the Compare view-model tests under a single-threaded synchronisation context like the UI thread (test-only), or record `_running` before clearing `_pending`; then re-run the assembly repeatedly

No screen-reader (Narrator / NVDA) session has been run. Not verified.

Risk table (illustrative, IEC 62304 §7 style)

Hazardous situation	Cause	Control	Verified by
Wrong position read off an image	y-flip or half-pixel error in screen ↔ mm	one mapping in `HeatmapViewport`; pixel-centre convention shared with the decoder	SR-VIEW-05 tests; VAL-03
Old image taken as current	scene edited after acquisition	physical inputs locked while data exist; Reset discards the data before an edit; Stop keeps acquired data with state text	SR-RUN-10, SR-RUN-12, SR-RUN-26 tests; desktop acquisition validation pending
ROI sum from the wrong image or frame	measurement not refreshed, or pane mixed up	measurements stored per pane in mm; refreshed on every result	SR-MEAS-04 tests
False precision	sub-pixel ROI weighting	whole pixels by centre; values at 0.1 resolution	SR-MEAS-03 tests
App hangs or crashes on a long or failing acquisition	MC on the UI thread; unhandled exception	worker transport / decode; cooperative Stop; failure → `State = Failed`	SR-RUN-03, SR-RUN-09, SR-RUN-10 tests
Status missed by colour-blind or screen-reader users	colour-only cues	text next to every status dot; selection by shape; live status line	SR-A11Y-04 (I); AN-06

7. Configuration and regression

Re-run verification

```
dotnet build Gcam.sln -c Release                     # all projects incl. the WPF shell
dotnet test  Gcam.sln -c Release                     # engine + Studio
dotnet test  tests/Gcam.Studio.Tests -c Release      # Studio logic only, no UI stack needed
dotnet test  tests/Gcam.Studio.Services.Tests -c Release   # the service against the real engine
dotnet run   --project src/Gcam.Studio -c Release    # for the §5 scenarios
```

A verification record is valid for one working tree: note the commit, the SDK version and the pass counts when the matrix in §4 is updated.

Rules that keep this document true

- Test names follow `Subject_ExpectedBehaviour[_Condition]`, one class per type under test, so a matrix row names one method.
- A change that adds or alters behaviour adds its requirement row (SRS (GCAM-SW-SRS)), its unit allocation (SDS §7 (GCAM-SW-SDS, 7. Requirement allocation)), its matrix row here and its test in the **same commit**; docs and code never drift apart.
- Renaming a test means updating its row here; a cited test that no longer exists is a defect in this document.
- Problems found during V&V go to §6 with an `AN-` ID and stay there until fixed.

Dated verification records

The dated matrices and passes (historical test inventory 2026-10-01, Waveform and shared-chain matrix, Detector and retained-flood focus matrix, final desktop verification, UI polish verification) are kept in VV.Studio.History (GCAM-SW-STR); the current state is this document.

Annex A — VV.Studio.Acquisition — live list-mode verification and measurements

Scope: engine list-mode events and the Studio acquisition contract, state and imaging snapshots. This is the acquisition-only verification baseline; Spectrum evidence and current test totals are in VV.Studio. The original baseline below used no desktop; subsequent acquisition desktop validation is recorded in VV.Studio (GCAM-SW-STR, Final desktop verification and validation (2026-10-02)).

At a glance

- Fresh histories, importance-weight rejection and independent Poisson timing feed one cumulative event list.
- Current realism and background evidence is below; original list-mode measurements remain the baseline.
- Current normal-run inventory is in VV.Studio; historical run counts remain below.
- All original 246 engine assertions remain unchanged. The unused batch API, commands and batch-only tests have been removed.
- Acquisition requirements extend through SR-RUN-29 (the absolute ambient field); SR-RUN-14 and the affected batch rows are withdrawn. Continue / Reset replace stale marking.
- Desktop migration history is below; the final four-workspace pass verifies current acquisition control.

Physics verification

Measured 2026-10-01, Release, seeded configurations loaded from the lab sample. Independent references use cloned configs with changed seeds. The weighted flood reference is the Compton Argmax pipeline accepting all deposits; the ideal geometric detector is not a crystal-deposit reference.

Test	Result	Tolerance / conditions
`PixelHistogram_MatchesIndependentWeightedComptonFlood`	maximum absolute z = 3.889 over 144 pixels	5σ per fraction, allowing 144 simultaneous comparisons; 100,000 physical events vs 1,000,000 reference histories; reference uncertainty bounded by $w^2 \le w_{max} \cdot w$
`MixedDistances_RejectionPreservesWeightedFloodAndTotalEmissionRate`	4.8077% rejection acceptance; max pixel z = 2.467; running rate 166.131 vs 165.774 cps	5σ per pixel; 4σ rate tolerance 5.634 cps; 40,000 physical events vs 2,000,000 reference histories; 160 mm / 1000 mm source distances, activity 1:40 — stresses unequal weights
`Deposits_MatchIndependentWeightResampledEventStream`	maximum absolute z = 1.199 over eight energy bins	4σ including reference uncertainty; 100,000 events vs 150,000 weight-resampled deposits; conservative reference effective size n/2 for these nearly equal weights
`RateAndGaps_MatchPhysicalEfficiencyAndExponentialLaw`	79.870 cps vs independent weighted MC 80.236 cps	4σ = 1.278 cps; last 90,000 of 100,000 events, excluding startup rate-estimator transients
Same test, normalized gap mean	0.99837 vs 1	$4/\sqrt{50{,}000} = 0.01789$; gaps multiplied by the running rate actually used
Same test, normalized gap variance	0.99713 vs 1	$4\sqrt{(8/50{,}000)} = 0.05060$, using the Exp(1) fourth central moment
Same test, gap CDF at 1	0.63574 vs $1-\exp(-1) = 0.63212$	4 binomial standard errors = 0.00863

Test	Result	Tolerance / conditions
Same test, unique events	100,000 distinct records; strictly increasing arrival times	exact; producer emits each accepted history once, with no deposit pool
`SeedAndCancellation_AreDeterministicAndPrompt`	same seed → identical 1,000-event sequences; pre-cancelled Advance throws	exact; cancellation checked every history
`PixelSink_PreservesLegacyEnergySinkAndTransport_WithOpticalCrosstalk`	old energy sink, richer energy sink and pixel deposits identical; flood unchanged	exact, shared photons and separately seeded detector cascades; optical crosstalk 0.2
`ShortAcquisition_LocalizesAfterCountThreshold_SnapshotsAreImmutable`	450 counts in 5 live s; localization error 2.158 mm	≥50 counts, error below one resolution element 8.750 mm; Cs-137 500 µCi at (15,8) mm, distance 1 m, default Studio optics

The running-rate timing is intentionally adaptive as specified: its finite-startup rate estimate is not a known exact constant. The gap tests normalize by that conditional rate, and the physical-rate comparison uses the mature prefix. No tolerance was loosened.

Throughput and reconstruction

Machine conditions: Intel64 family 6 model 183 stepping 1, 24 logical processors, Windows build 26200, .NET SDK 9.0.311 / runtime 9.0.13. Accepted throughput is physical events per measured wall second, including MC generation, transport and rejection. It excludes reconstruction and UI drawing. One producer thread; 2,000-event warmup followed by 100,000 events. Twenty decoder calls follow one decoder warmup.

Configuration	Accepted events / wall s	Rejection acceptance (accepted / detected histories)	Accepted / all proposals	Decode mean / max
`samples/scenario.json`, seed 12345	438,731 (100,000 / 0.2279 s)	99.8861% (100,000 / 100,114)	21.1848% (472,036 emitted)	4.585 / 4.917 ms
`samples/scenario_handheld.json` cloned; source–mask distance = 945 mm, detector distance = 1 m, Cs-137, seed 12345	330,918 (100,000 / 0.3022 s)	99.9950% (100,000 / 100,005)	27.4590% (364,180 emitted)	8.598 / 12.031 ms

The default Studio grid has more detector pixels than these sample rigs. A real service refresh in the short acquisition test decoded in 28.907 ms (includes first-use/JIT effects); still below the 250 ms refresh interval. The service budgets transport to 200 ms per refresh, then decodes the cumulative flood. These are local CPU measurements, not desktop rendering performance or guaranteed throughput. Large event lists are kept for future spectrum reprocessing; detached snapshot copies have memory / copy cost proportional to acquired counts.

Reproduce without desktop UI:

```
dotnet build Gcam.sln -c Release -m:1 -nr:false
dotnet test Gcam.sln -c Release -m:1 -nr:false
dotnet test tests/Gcam.Tests -c Release --no-build --filter FullyQualifiedName~ListModeSourceTests --
logger "console;verbosity=detailed"
dotnet test tests/Gcam.Studio.Services.Tests -c Release --filter
FullyQualifiedName~AcquisitionServiceTests --logger "console;verbosity=detailed"
```

Leave `GCAM_UI_TESTS` unset. Build has zero errors; NU1900 reports unreachable NuGet vulnerability metadata. Core's 69 cases include concurrent additions to the existing plot tests and four acquisition cases.

Requirement and design trace

VV.Studio.SRS (GCAM-SW-SRS): SR-RUN-01, -02, -04, -05, -06, -07, -08 retain their IDs marked withdrawn. SR-RUN-03 (failure handling) remains active. SR-RUN-09 … -19 cover background acquisition, Stop retention, live-time progress, input locking, Start availability, input validation, immutable shared events, 4 Hz decode / measurement refresh, MC limitation and preset completion; SR-RUN-14 (stale marking) is withdrawn and SR-RUN-23 … -28 add Continue, the raised preset, the seed, Reset, failure handling and the observed status rate; SR-RUN-29 adds the absolute ambient field. The matrix is in VV.Studio; SU-19 is added in VV.Studio.SDS (GCAM-SW-SDS).

Virtual-clock tests verify snapshots grow, preset completion, Stop retention, Continue, Reset, measurement geometry and input locking. Service tests verify real transport/localization, detached immutable snapshots, count/flood conservation, flood pixel centres matching the decoder to nine decimal places, look-ahead boundaries, input rejection and an extreme MC-limited Stop on an injected clock. Read-only cross-review identified and rechecked fixes for Stop's live horizon and true deposits before optical light spread. No desktop session was started.

The batch progress implementation and its late-callback regression test have been removed with the unused batch path. Acquisition progress comes from sequential snapshots.

UI-test migration (applied 2026-10-01)

Historical migration record: the later acquisition-control change withdraws source drag and stale marking. Continue now preserves data; Reset clears them. Final desktop evidence is in VV.Studio.

The desktop scenarios now drive Start / Stop / live time: `StudioWindow.Acquire` waits for Completed / Stopped / Failed and the status-line counts are parsed; the cancel scenario became `Stop_KeepsAcquiredData_LocksThenUnlocksScene` (inputs locked while acquiring, counts frozen after Stop, Start clears). On the desktop all 19 UI tests passed and, with corrupted verdicts, all 7 desktop scenarios failed at their corrupted assertions. Not yet automated: preset auto-completion timing, count monotonicity during acquisition and the MC-limited presentation.

Scope

Studio uses Start / Stop / Continue / Reset. The batch members, ISimulationService and batch enum values have been removed; SimulationService remains as the acquisition adapter and ImagingResult remains the snapshot image record. Before list-mode acquisition the flood used geometric CrystalDetector scoring through DefaultSimulationFactory. It now adds one count per ComptonCrystalDetector event at its Argmax pixel, so in-crystal Compton scatter mispositioning is included in the image. No quantitative size of this change is asserted.

Studio supplies detector settings explicitly after cloning the scene config; existing builder callers keep the bare geometry. List-mode deposits remain unsmeared and unwindowed, located by Argmax. Measurement applies the seeded CrystalUniformity gain pattern before one chain smear. Snapshots retain their acquisition inputs, and the gain inputs are locked while data exist, so the recorded response cannot change. Spectrum windows and optional pile-up reuse these events. Per-nuclide imaging and stripping are outside this scope. A Co-60 or Na-22 decay is one event: the summed deposit of its detected gammas at the largest-deposit pixel over their merged sites, one arrival time.

Detector realism and background

Measured 2026-10-01, Release. Source tests use default Studio optics (30×30 pixels, 0.6 mm pitch, Cs-137 at 1 m), transport seed 12345, measurement seed 909, gain σ 3% and gain seed 1. Entrance is 0.15 mm steel-equivalent, backing 2 mm and reflector gap 0.1 mm. Each absorber/backing comparison transports 2,000,000 proposals per configuration; variations use Clone + mutate. Counts are per proposal budget, not normalized to a fixed accepted event count, so attenuation remains visible. The spectrum has a fixed 0–2000 keV axis of 1000 bins, 2 keV wide. The peak-bin ratio and the histogram FWHM below are re-measured on these bins (2026-10-04); the grouped Ba band and the 480–620 keV valley counts are the record on the earlier 2.9725 keV bins, and their tests pass unchanged on the new ones.

Re-measured 2026-10-02 after the random generator was replaced (xoshiro256**, per-event smear streams keyed by (seed, index)): every row below is the new seeded realisation; all assertions pass with the unchanged tolerances. The throughput table above keeps its 2026-10-01 timings; its seeded acceptance / proposal counts are re-measured.

Test / check	Measurement	Expectation and tolerance
`BaKAbsorber_UncollidedBandMatchesIndependentNarrowBeam_ReportsMeasuredSpectrum`, unchanged Ba line energies	50,064 → 23,777; ratio 0.474932	Beer-Lambert prediction 0.474899 ± 0.014987 (4σ), weighted by the bare 32.1 / 36.4 keV line counts; independent log-log interpolation of iron μ anchors at 30 / 40 keV gives transmission 0.450696 / 0.571939; Poisson reference and transmitted-count uncertainty included
Same test, measured grouped Ba band	57,718 → 35,736; ratio 0.619148	independent weighted transport + measurement MC, seed 987 / 444, predicts 0.619379 ± 0.028971 (4σ); weighted-reference variance bounded by $w^2 \le w_{max} \cdot w$
Same test, Ba K / 662 peak-bin ratio	4.187313 without absorber, 2.320998 with absorber	descriptive, not a global-peak requirement; Ba K remains the global maximum at 33.0 keV
Same test, 480–620 keV valley (P-11)	11,157 → 11,559 counts	descriptive; increase 402 counts is 2.7σ under an independent Poisson comparison (was 148 counts, 0.98σ, with the old generator); no claim of a large absorber tail
`Backing_AddsBackscatterRegion_FromTransport`	170–210 keV counts 15,415 without backing → 17,578 with backing; excess 2,163	excess exceeds 4σ = 726.5590 counts; Compton's 180° return energy $E/(1+2E/m_ec^2)$ = 184.3263 keV; finite-angle return broadens toward higher energies; the bare crystal already has continuum in this region
`Gain_WidensPhotopeakInQuadrature_AndReducesTightWindowAcceptance`, 53,690 MC full-energy events	measured variance 560.0673 vs expected 557.3291 keV²; equivalent FWHM 55.7281 vs 55.5917 keV; sampled pattern σ 2.9932%	pixel-amplitude variance + mean FrontEndModel noise variance; 4σ variance tolerance 13.7996 keV² from the mixture's fourth central moment
Same test, histogram FWHM	54.5845 vs 55.5917 keV	quadrature prediction, tolerance 4.8482 keV = two bins + 5·FWHM/√(2(N−1))
Same test, ±0.5 chain-FWHM window	fraction 0.761836 at zero gain spread → 0.477500 at 3%; decline 0.284336	decline exceeds 4σ = 0.011332, conservative independent-binomial bound for paired samples
`BsrOne_DoublesRate_PreservesEverySourceEvent_AndMatchesBackgroundModel`	79,943 source + 79,825 background events in 1000 s; mature total/source ratio 1.997237	BSR=1 predicts 2; 4σ tolerance 0.021062, excluding first 100 s of running-rate startup
Same test, spatial / energy distributions	uniform-profile max abs z 3.0968 over 144 pixels; energy max abs z 1.3247 over eight bins	5σ per pixel vs Background.SideLeakProfile(sideFraction=0); 4σ per energy bin vs 100,000 fresh cosine-flux deposits from BackgroundDepositSpectrum at 200 keV with independent seed 987
`ZeroBsr_PreservesSourceStreamBitForBit`, `Background_SeedAndCancellationAreDeterministic`	source-only and explicit zero-BSR records, histories and weights agree; enabled streams replay; cancellation throws promptly	exact seeded equality, no extra RNG draws on the disabled path

Test / check	Measurement	Expectation and tolerance
`Measurement_UsesFrozenInputs_AndReplaysAcrossSnapshotsAndPileUp`	batch, incremental and toggle replay agree; shared MeasurementStage reproduces every histogram bin	exact; 8,427 fresh events; gain precedes smearing and is not applied twice

Narrow-beam transmission describes uncollided photons, not the full measured Ba band: entrance Compton scattering can leave a photon inside that broad band. Both checks are therefore reported separately. The absorber suppresses Ba counts, but in this geometry does not make the Ba peak smaller than the 662 keV peak. No tail, line or noise floor is added by hand. The valley increase is small and statistically unresolved here.

The BSR background's energy response matches the existing unmasked cosine-flux crystal model; pixel placement matches the uniform detected pedestal. BSR is defined after detection, so this is not a prediction of incident flux, entrance loss or shield leakage. Each background deposit is freshly transported and used once; no event pool is repeated. Source records remain exactly intact when background is merged. Source and background arrival streams use separate RNGs and the running source-rate estimator. Spatial structure beyond the uniform pedestal is outside this implementation.

Absolute ambient field (default). Studio starts with 0.10 µSv/h photon H*(10), validated terrestrial spectrum, front-only bound (SR-RUN-29). In the startup scene (Cs-137 500 µCi at 1 m, default optics and detector, 60 s) the field adds 0.137 ± 0.008 cps (4 × 600 s source-free) against 73 cps from the source — 0.19 % of the counts, 0.03 % of the 662 keV window; localisation is unchanged within the seed scatter (8 seeds). The bare-crystal bound gives 30.4 ± 0.1 cps (29 %). On the fixed 0–2000 keV axis the 662 keV band count is the same with or without the field's line list (1349 ideal, 1354 default, 8 seeds); the default field adds ≈ 0.25 overflow counts per minute. These are headless measurements through the Studio services; the desktop scenarios have not been run with the field on.

The first background test used a reference-only variance estimate, then incorrectly treated an empty finite reference bin as zero probability. The two-sample estimate now pools both counts under the equal-distribution null; bins with no variance in either sample require exact agreement. The 4σ energy criterion is unchanged.

Throughput with realism defaults

`Throughput_RecordsBareAndRealisticAcquisition` warms each producer for 2,000 accepted events and times 100,000 further events, four runs per configuration. Bare Studio scene-builder geometry gives 536,360, 525,671, 538,449 and 538,924 accepted events/s (mean 534,851). Realism defaults give 285,234, 299,889, 291,842 and 284,532 events/s (mean 290,375): about 45.7% lower accepted throughput. This comparison includes transport and rejection, excludes measurement / decode / UI, and changes physical geometry as well as its compute cost; it is a local measurement, not a performance requirement.

Traceability and pending desktop checks (historical implementation record)

Current status: the final desktop pass covers workspace clicks, detector locks, retained Spectrum windows and Imaging/focus projections. SelectionItemPattern and keyboard-only/screen-reader coverage remain separate gaps; see VV.Studio.

SR-RUN-20 … -22 cover explicit defaults, frozen measurement inputs and background. SR-NAV-01 / -02, SR-SPEC-02 / -03 and SR-RUN-12 state the shared inputs, checked-state workspace activation, gained amplitudes and input locking. Headless workspace activation fixes AN-10; SelectionItemPattern.Select still needs a real desktop regression. No Studio window was launched and the UI-test project was not edited.

Needed UI additions: select both workspaces through SelectionItemPattern and assert the displayed content; verify detector defaults, field reachability in both themes at minimum size, gain / seed / BSR locking during acquisition and while data exist, preserved spectrum response after a later window change, and live BSR count/rate presentation. Existing desktop tolerances remain unchanged.

Start / Stop / Continue / Reset (2026-10-02)

The acquisition follows the multichannel-analyser model: Start acquires, Stop pauses keeping everything, Start again (labelled Continue) resumes the same acquisition, Reset discards it. Physical inputs are locked while data exist; view settings are not. Each new acquisition uses a new Monte Carlo seed unless one is fixed.

Continuation is exact. The session keeps the list-mode source with all its random streams, the events, flood, decoder, live-time clock and the one event already drawn beyond the acquired live time. A stopped and continued acquisition therefore produces the uninterrupted event stream, event for event. Measured on the real session with an injected clock (`AcquisitionContinuationTests`, Cs-137 500 µCi at (0, 0) + Co-60 300 µCi at (20, 0) mm, 1 m, seed 4711, preset 12 s, four stops at live 1, 2.125, 7.625 and 8.5 s at speeds ×4, ×1.5, ×11, ×0.7, ×6, 100 s of wall time while stopped each time):

Background	Events in 12 s	Result
BSR 0	1,907 (1,810 with the old generator)	identical to the uninterrupted session and to the direct source stream; arrival times strictly increasing; each continued segment starts at the stopped live time and counts
BSR 0.5	2,769 (old 2,728)	identical, as above
Co-60 500 µCi at (3, −2) mm, 100 mm, BSR 0 (same stops)	139,933 (62 decays with both gammas detected; old 139,747 / 48)	identical, as above; one event per decay keeps the single look-ahead event sufficient

A raised preset after Completed (3 s → 6 s) continues identically to an uninterrupted 6 s acquisition; the same seed reproduces an acquisition and the next seed gives an independent one. Sensitivity check: with the look-ahead event deliberately dropped at each Stop, all three continuation tests fail at the first stop (one real count lost per stop). Wall-clock pacing never enters the events: speed only decides when the worker consumes them, and live time, not wall time, is the physical clock, so it does not advance while stopped. The Poisson arrival process is continued, not restarted: the next arrival is the one drawn after the last event.

Not verified here: the desktop (Reset placement, keyboard order, Continue label) — the rewritten desktop scenarios are not run.

Annex B — VV.Studio.Imaging — per-nuclide imaging verification evidence

Scope: measured-energy imaging channels and Compton stripping in GCAM Studio. This page separates association verification from measured localization precision; it makes no new precision claim.

At a glance

- Requirements SR-IMG-01 … SR-IMG-06 are in VV.Studio.SRS (GCAM-SW-SRS).
- Headless requirements passed; Release measurements below were supplied by the reviewer on 2026-10-02.
- The Co-60 y bias near −1.4 mm is not an energy effect: it is a position-dependent localisation error from undersampling the mask-cell shadow with the default optics at a 1 m focal plane (see "Localisation bias").
- Retained offscreen mixed-scene renders cover both themes and sizes. The final desktop pass verifies channel selection, stripping and bound found peaks; see VV.Studio.
- The measurements below are headless; the later desktop pass is separate evidence, not a new localisation-precision measurement.

Current normal-run totals are in VV.Studio. Dated verification records below retain the execution limits at that time; subsequent desktop evidence supersedes their pending status.

Scenes and checks

Defaults: rank 13, mask pitch 0.7 mm, D 80 mm, focal/source distance 1000 mm, 30×30 pixels at 0.6 mm, Studio detector realism (entrance 0.15 mm, backing 2 mm, reflector gap 0.1 mm, gain σ 3%, seed 1), default physical front-end chain, N = 1.5. Cs activity 500 µCi; Co activity 1000 µCi. All events come from ListModeSource; the measured response is MeasurementStage.

Check	Method and acceptance	Measured result
Co-located Cs + Co at (0,0), 600 s	Raw low window must exceed an independent same-live-time Cs-only reference by $4\sqrt{(L+C)}$. Strip count acceptance is $4\sqrt{(L + R^2H + C + H^2\mathrm{Var}(R))}$; $\mathrm{Var}(R)=R^2(1/L_{cal}+1/H_{cal})$, delta-method independent Poisson calibration windows. Seeds 27182 mixed and 31415 Cs-only.	Passed. R(Co→Cs) = 0.4706 (6,018/12,787). Raw Cs L = 23,009, high H = 20,410, stripped Cs = 13,421.7843, Cs-only C = 13,331; error = 90.7843 counts versus 4σ = 1,007.2427 counts; Var(R) = 5.4128E-05. (Old generator: R 0.4775, error 112.0634 vs 4σ 1,016.5072.)
Off-axis association, 600 s	Cs (15,8) mm, Co (−15,−8) mm, seed 19283. Each refined top peak error must be strictly below step × √2 = 3.093592 mm (step 2.1875 mm). This is an association bound, not a precision tolerance.	Passed. Cs found (15.2480, 7.8387) mm, error 0.2959 mm; Co found (−15.3164, −9.3830) mm, error 1.4188 mm. Both below the 3.0936 mm association bound. R(Co→Cs) = 0.4709 (6,020/12,785). (Old generator: errors 0.1765 / 1.4238 mm, R 0.4825.)
Localization precision	20 seeds 42000 … 42019 per scene, 600 s each. Mixed Cs / Co and Co-only control at the same Co position/activity. Record mean error vector, RMS radial error and maximum radial error, raw and Tent-refined. No precision assertion or tuning.	Completed; measurements in the table below. Co-60 y bias also appears without Cs.
Window replay	Seed 13579, 60 s mixed co-located acquisition. N 1.5 → 2; retained prefix matches fresh processing exactly, R recomputes, strip-only replay uses cached R.	Passed. At N = 2, R(Co→Cs) = 0.5269 (8,038/15,255), recomputed (old generator 0.5336). Strip toggle timings below are from the 2026-10-01 run: channels 0.299 ms, calibration 0 ms, decode 37.7 ms (co-located 600 s check).

Calibration accepts 100,000 H-only events (above the required 20,000). The source's complete emission lines, geometry, gain and detector response are preserved. Background is disabled for H-only calibration. The primary Co window is 1173.2 keV; calibration includes its 1332.5 keV emission. R and channel / calibration / decoding milliseconds are emitted by each test and displayed in the panel. For the off-axis check, 100,000 H events were accepted; worker costs were channels 216.2 ms, calibration 696.8 ms and decode 90.7 ms.

Scene / isotope	Peak method	Mean error (x, y), mm	RMS radial error, mm	Maximum radial error, mm
Mixed / Cs-137	Raw grid peak	(−0.125, −0.781)	1.350	1.448
Mixed / Cs-137	Tent-refined	(−0.080, −0.115)	0.234	0.418
Mixed / Co-60	Raw grid peak	(0.344, −2.281)	2.307	2.307
Mixed / Co-60	Tent-refined	(−0.208, −1.368)	1.387	1.483
Co-only / Co-60	Raw grid peak	(0.344, −2.281)	2.307	2.307
Co-only / Co-60	Tent-refined	(−0.199, −1.368)	1.387	1.481

Re-measured 2026-10-02 after the random generator was replaced (xoshiro256** with independent per-event smear streams). Earlier values (old generator): Cs raw (−0.453, −0.781) / 1.380 / 1.448; Cs refined (−0.093, −0.084) / 0.167 / 0.227; mixed Co refined (−0.194, −1.378) / 1.395 / 1.456; Co-only refined (−0.200, −1.367) / 1.384 / 1.465. The Co-60 y bias is unchanged (−1.378 → −1.368 mm). The Cs refined RMS rose 0.167 → 0.234 mm — about 1.5 standard errors of a 20-sample RMS, so not established as a change; the old generator smoothed the per-event energy smear (window counts fluctuated less than binomially), which acts in this direction.

The engine EV-15 scalar subtraction model motivates this method, but does not substitute for these Studio measurements. Finding TODO-17: at Co-60 (−15,−8) mm, 1000 µCi, with the defaults above, N = 1.5, stripping enabled, 600 s and 20 seeds, the refined mean y error is −1.368 mm in the mixed scene and −1.368 mm in the Co-only control. This records a systematic bias under these conditions; its cause is given under "Localisation bias" below. The Cs refined mean error is (−0.080, −0.115) mm in this mixed-scene sample. These numbers do not establish precision at other positions, geometries, activities or live times.

Localisation bias: detector sampling, not energy (2026-10-02)

Single-source, weighted engine runs (3 × 10^6 biased photons, noise negligible; default optics, focal 1000 mm): the same bias appears with the ideal geometric detector and with Compton Argmax events, for Cs-137 as for Co-60 (at (−15, −8): Cs −1.26 mm, Co −1.40 mm in y), and it depends on the source position (y = −8: −1.3 … −1.4 mm at any x; y = +8: −0.3 mm; y = 0: +0.15 mm). A 0.25 mm reconstruction grid does not remove it, so it is not the sub-cell interpolation. Varying the detector pixel pitch at constant detector size (Cs-137, x = 0, y = −10 … 10 mm in 1 mm steps, 0.25 mm grid):

Pixel pitch	Samples per mask-cell shadow (0.761 mm)	RMS error	Max error
0.6 mm (default)	1.27	0.95 mm	1.91 mm
0.3 mm	2.54	0.51 mm	0.91 mm
0.2 mm	3.80	0.24 mm	0.41 mm

The error repeats with the source position (period ≈ 3.4 mm at 0.6 mm pitch): the shadow's phase on the pixel grid decides where the decoded peak lands. The default optics reach ≥ 2 samples per cell only near a 160 mm focal plane (shadow 1.4 mm), not at 1 m.

Reproduction and renders

Run `dotnet build Gcam.sln -c Release`, then `dotnet test Gcam.sln -c Release` with GCAM_UI_TESTS unset. The 20-seed precision measurement and the full sampling regression are now long-running evidence tests, skipped by default and in CI. Their budgets and historical measured values above are unchanged. The full automated sampling sweep uses 1,000,000 biased histories per position, pitches 0.6/0.3 mm, y=-10..10 in 1 mm steps, x=0, a geometric Cs-137 detector at 1 m, fixed 18 mm detector size, seed 12345 and 0.25 mm grid. It measured RMS/max 1.0734/1.9966 mm and 0.5847/0.9525 mm (old generator: 1.0810/1.9966 and 0.5837/0.9511); this is distinct from the original 3,000,000-history, three-pitch table above. Reproduce these automated measurements with the independent evidence opt-in:

```
$env:GCAM_EVIDENCE_TESTS = '1'
dotnet test tests/Gcam.Studio.Services.Tests -c Release --filter Category=Evidence --logger
'console;verbosity=detailed'
Remove-Item Env:GCAM_EVIDENCE_TESTS
```

`Sampling_PositionSweepAtConstantDetectorSize_FinerPixelsReduceLocalizationError` preserves the full position sweep; `Precision_TwentySeeds_MeasuresMixedIsotopesAndCoOnlyControl` preserves the 20-seed measurement. `Sampling_SmallBudgetSeedSpread_ReportsRegressionMargin` measures noise in the separate small default regression budget. None of these opts into desktop input.

The default sampling regression pins seed 12345, 100,000 biased histories per position at x=0, y={-8,-4,4,8} mm, geometric Cs-137 at 1 m and an 18 mm detector (30×0.6 / 60×0.3 mm). Its assertion is fine-pixel RMS < 0.85 × coarse-pixel RMS. The reduced photon/position budget was measured with five seeds using the original full-field 0.25 mm grid, independently of the high-budget 0.95/0.51 mm finding:

Seed	Coarse RMS, mm	Fine RMS, mm	Fine / coarse
12345	1.109679	0.631894	0.569438
23456	1.108024	0.622447	0.561763
34567	1.068757	0.631782	0.591137
45678	1.124369	0.520752	0.463150
56789	1.110107	0.632080	0.569386

The observed ratio range is 0.127987; max + twice that range is 0.847111, rounded up to 0.85. This gives 0.258863 ratio headroom above the worst sampled seed. It is an empirical margin for a pinned deterministic test, not a confidence bound or a guarantee for arbitrary seeds.

Re-measured 2026-10-02 with the replaced random generator on the final cropped grid (the opt-in `Sampling_SmallBudgetSeedSpread_ReportsRegressionMargin`): fine / coarse ratios 0.528149 (12345), 0.523767 (23456), 0.506160 (34567), 0.522627 (45678), 0.433514 (56789) — range 0.094635; max + twice the range = 0.717, inside the 0.85 gate (the gate is unchanged).

The calibration run above took 193.9 s: photon reduction alone did not remove full-grid decoding cost. The final small regression therefore uses half-extent 12.125 mm and step 0.25 mm (98×98), retaining the nominal half-field 56.875 mm grid's phase. All four source positions are inside this region; every sampled radial error is at most twice its four-position RMS (≤2.249 mm), so the measured peaks and their local interpolation neighbours lie inside it. Transport/flood scoring is unchanged. The same budget's seed-spread opt-in now uses this small grid too. Its final equivalence, runtime (<10 s target) and default service-suite duration could not be checked after a shell command was blocked with `CreateProcessAsUserW failed: 5 (access denied)`; the table records the full-grid calibration, not a completed execution of the final cropped-grid test. Set GCAM_RENDER_SNAPSHOTS=1 only for `tests/Gcam.Studio.RenderTests`. Never enable desktop tests for this check.

Generated additional snapshot paths: `docs/assets/studio-render/imaging-mixed-{all,cs-137}-{dark,light}-{1280x800,1440x900}.png`. These use analytic drawing fixtures to exercise the production selector, selected image binding and diamond overlay. Physics evidence comes exclusively from the real-engine tests above.

The pre-fix mixed render failed at the channel-items assertion before the dispatcher binding flush. Instrumentation of `BindingOperations.GetBindingExpression` showed the options panel's ancestor DataContext binding was Active, with a `System.Windows.Controls.Border` source whose DataContext was `Gcam.Studio.Core.ViewModels.ImagingWorkspaceViewModel`. However, the ComboBox ItemsSource binding (`Isotopes`) had status Unattached and a null DataItem at that point. The observed failure was deferred attachment of the child binding, not a demonstrated lookup of the wrong ViewModel.

The options panel now inherits the workspace directly. Only its sibling measurement section binds to `Measurements`; the Border ancestor lookup is removed. At the same pre-flush assertion, ItemsSource now has status Active and DataItem equal to the workspace instance. The render test checks these identities, the All / Cs-137 / Co-60 items, selection, image identity and found-peak count.

MLEM reconstruction at the default optics (TODO-36, 2026-10-07)

Conditional evidence for SR-IMG-07 at Studio's default optics (rank 13, 0.7 mm cells, D = 80 mm, 30 × 30 pixels at 0.6 mm, focal plane 1000 mm; transported maps, 662 keV window, exact Poisson, the blind pair test). Not claimed for other optics or focal planes; the UI says so.

- **Iteration count.** The pair-resolution rule (smallest resolved separation with single-source false splits ≤ 5 %; ties to fewer iterations) picked 360 on selection seeds (15 × 50); on check seeds (16 × 50) 360 passed 1.0 element at 0.949 [0.927, 0.968], so 400 was taken and confirmed on a third, disjoint set (16 × 50): **1.0 element 0.960 [0.940, 0.979]** (point estimate against 95 %), false split ≤ 0.13 % [0, 0.37] at 2000 counts; cross-correlation resolves 2.0 elements. Cost at low counts: a 250-count single source shows a second peak ≥ ¼ of the main height in 50.4 % [46.3, 54.2] of frames (≥ ½: 11.6 %); cross-correlation 100 % / 77.5 %.
- **MLEM + strip on Cs-137 under Co-60** (16 seeds × 50, Studio's windows; 360 iterations, ideal strip ratio). Cs position RMS / count bias (RMSE), cross-correlation + clipped strip against MLEM with the downscatter as background: 1000 Cs counts, Co : Cs 1 : 1 — 0.071° / +88 (95) against 0.043° / +1 (39); 4 : 1 — 0.132° / +249 (253) against 0.120° / 0 (54); 4000 counts, 1 : 1 — 0.055° / +45 (89) against 0.025° / −3 (80); 4 : 1 — 0.064° / +222 (241) against 0.030° / −1 (104). The clipped strip sum read 4.5–25 % high; since TODO-39 both methods report the signed net.
- **Display rate** (real session, 5 channels, 60 s at speed 10): cross-correlation 3.4–3.6 Hz (worker decode 80–90 ms median); MLEM at 360 iterations 1.1–1.3 Hz (600–740 ms); about 1.0–1.2 Hz at 400 (scaled, not re-measured). The producer held 9.9–10× speed; transport never stalled.

Strip count and signed decoding (TODO-39, 2026-10-07)

Conditional evidence for SR-IMG-05 at the default optics (16 fresh seeds × 50 per Cs-under-Co scene). The signed net Σ low − R·Σ high is unbiased within −3.4 … +1.6 counts, where the old clipped sum read +50 … +250 high; empirical SD / reported σ 0.94–1.04 (two seed-bootstrap intervals exclude 1, so exact agreement is not claimed). Decoding the signed difference instead of the clipped flood: cross-correlation RMS 0.039–0.073° against 0.039–0.083°, one peak per frame, no selected peak outside one element or nearer Co-60. An independent statistical test (4096 draws, bounds fixed from the distribution's moments before sampling): mean −1002.61 against −1002.51 (± 8.17), reported σ 130.7 against an empirical 129.3 counts. Limits: scalar strip model, one contaminant for the reported σ, no ambient background removal.

Verification status

Reviewer verification (Release, 2026-10-02): build 0 errors; engine 259, Studio Core 86, services 28 and UI oracle 10 tests all passed. Numerical results above are the reviewer's measured ImagingServiceTests output, not newly measured timings from the render fixture. Local verification and snapshot inspection status are recorded in VV.Studio. Desktop validation remains pending; no desktop UI tests or Studio launch were performed.

Annex C — VV.Studio.Detector — detector geometry and retained-flood focus verification

Scope: editable reflector gap, the static Detector workspace and experimental focus analysis in GCAM Studio.

At a glance

- Detector displays a seeded relative-gain pattern and exact geometric dead gaps, with acquired and next-acquisition inputs separated.
- Focus projects retained channel floods; it generates no transport histories and has no calibrated depth accuracy.
- Half-maximum intervals are descriptive and censored at sweep edges; an external surface range is a separate observation.
- Deterministic tests run normally; long numerical evidence, offscreen renders and desktop tests require their separate opt-in variables. Current totals and retained desktop evidence are in VV.Studio.

The final desktop pass checks the face before acquisition, acquired-input locks through Reset, refocus projection and raw half-max endpoints. Earlier execution restrictions below are historical; they do not describe current workspace availability. See VV.Studio.

Conditions and numerical expectations

Inherited Studio-path measurements used rank 13, mask cell 0.7 mm, D=80 mm, 30×30 crystals at pitch 0.6 mm, GAGG, Cs-137 500 µCi, on-axis detector-referenced z=1000 mm, 60 s per seed, physical entrance/backing, gain σ=3%, seed 1, and a 661.7-keV window ±1.5 chain FWHM (chain FWHM approximately 30.2 keV). Three transport seeds measured total accepted rate 103.70/72.81/46.63 cps for gaps 0/100/200 µm, or ratios 1/0.702/0.450. The independent analytic geometric expectations are 1/0.694444444/0.444444444. These are conditional rate comparisons, not a general equality between count efficiency and area. The primary-window fraction was about 0.30. These inherited values are context, not assertions or tolerances in the new test, and no new MC result is claimed here.

`DetectorGapEvidenceTests.RateRatio_MatchesGeometricArea_WithinMeasuredSeedSpread` reproduces the full Studio list-mode and measurement path. Twelve independent transport seeds (43001+i×1009) run 60 s at each gap, with paired zero-gap references. For each gap, the first eight ratios measure sample standard deviation s with divisor 7; tolerance is **4·s·sqrt(1+1/8)**, a per-seed predictive spread bound including calibration mean estimation. The calibration mean and four independent holdout ratios are compared with the analytic area expectation. Every seed's counts, cps, primary-window counts, ratios, s and derived tolerance are printed. No fixed tolerance, borrowed precision or fallback bound is present. A failure is a measurement result to investigate; this test does not establish a Gaussian coverage probability or universal fill-factor law.

Measured 2026-10-02 (opt-in run, Release), old and replaced random generator: at 100 µm the calibration mean is 0.69428 (s 0.01672, tolerance 0.07094) with the old generator and **0.69846 (s 0.01472, tolerance 0.06247)** with xoshiro256**, against 0.69444; at 200 µm 0.45548 (s 0.01521, tolerance 0.06453) → **0.44753 (s 0.01074, tolerance 0.04557)**, against 0.44444. Both pass; the changes are within about one standard error of an eight-seed mean.

Inherited focus measurements used five seeds and 10/60 s at the same default optics, with near planes also sampled on a 10 mm grid. At true z=300/500 mm on axis, sharpest broadband planes were approximately 360/580–586 mm, with half-max intervals 230–380/350–730 mm. At 30 mrad, sharpest planes were about 280/460 mm. At z=700 mm, off-axis best plane was about 560 mm and the interval reached the far edge. From roughly 700 mm the far endpoint was censored; at 2 m best planes ranged 890–2900 mm across seeds. These measurements motivate the visible bias/bound note, not a depth acceptance tolerance. The new 81-plane inverse-z grid is a sampling policy, with deterministic endpoint/spacing tests; the prior grids and their exact best planes are not silently asserted for this grid. Width is never treated as uncertainty.

Verification and commands

Scope	Test / fixture	Execution record
Units, validation, snapshot and acquired (locked) face	DetectorWorkspaceTests; DetectorFaceTests; Face_GainsMatchMeasurementPattern; Config_RejectsGapAtServiceBoundary	Passed in Release, 2026-10-02
Plane sampling, exact half-max crossings, censored edges, modes and angle linking	FocusSweepMathTests	Passed in Release, 2026-10-02
Frozen identity, K, cancellation, snapshot consumption and obsolete-result rejection	FocusSweepViewModelTests; ImagingWorkspaceTests channel invalidation	Passed in Release, 2026-10-02
Retained flood, no events/truth, empty flood and cancellation	FocusSweepServiceTests	Passed in Release, 2026-10-02
Area/rate relation	DetectorGapEvidenceTests (Evidence)	Requires opt-in run; no measured s or numeric tolerance obtained in this turn
Both themes and sizes	RenderDetectorAndFocus: detector-before, detector-acquired, focus-near-resolved, focus-far-censored	Fixture prepared; 16 new PNGs not generated or inspected in this turn

The render curves are analytic drawing fixtures, not synthetic physics evidence. The face fixture uses the actual engine gain pattern. All fixtures detach MainWindow content, never Show/Run, create an HWND or send desktop input. The existing single opt-in render case owns the sole STA/Application lifetime.

```
dotnet build Gcam.sln -c Release
dotnet test Gcam.sln -c Release
$env:GCAM_EVIDENCE_TESTS = '1'
try {
    dotnet test tests/Gcam.Studio.Services.Tests -c Release --no-build --filter
FullyQualifiedName~DetectorGapEvidenceTests --logger 'console;verbosity=detailed'
} finally { $env:GCAM_EVIDENCE_TESTS = $null }
$env:GCAM_RENDER_SNAPSHOTS = '1'
try {
    dotnet test tests/Gcam.Studio.RenderTests -c Release --no-build --logger 'console;verbosity=detailed'
} finally { $env:GCAM_RENDER_SNAPSHOTS = $null }
```

After process execution was restored, the Release solution build passed and the final ordinary solution suite passed: engine 259, Studio Core 147, services 67 and UI oracles 10. Seven service evidence cases, nine desktop cases and the one render case were skipped. Records are under `.artifacts/todo11/tests/` with the `todo11-final` prefix. The final incremental build reports zero warnings/errors; the earlier rebuild reported two existing xUnit analyzer warnings. No desktop test or Studio launch occurred. Opt-in environment variables were absent; changing them requires the author's approval under the execution limits. The two opt-in command blocks above restore that absent process-local state; they have not been executed. Numeric MC tolerances and PNG inspection remain unverified. No acceptance bound has been loosened.

Limits

The engine still assigns events directly to crystals. Crosstalk and SiPM pitch are excluded until a light-sharing and four-channel position readout exists. Minimum-cost angular tracking can exchange identities for close or crossing rays; K requests candidates and does not prove K sources exist. A same-ray mixture is not separated by this curve tool. Missing-plane tracks are unresolved. Background and clipped stripped floods change the descriptive metric and cannot supply a Poisson likelihood interval. Decoder cancellation occurs between planes; its bounded internal decode is not interruptible. External surface range is not proof of gamma-source range.

Requirements: VV.Studio.SRS (GCAM-SW-SRS). Design: VV.Studio.SDS (GCAM-SW-SDS).

Annex D — VV.Studio.Waveform — scope and acquired-chain verification status

Scope: shared physical detection-chain selection and the array-wide Waveform workspace; requirement/test mapping, execution limits and headless reproduction. Four-channel position simulation is excluded; the CR-RC Q12 arithmetic and preset timing contract are covered here.

The initial verification inventory and sandbox history below describe the original workspace delivery; the CR-RC arithmetic/response execution record at the end supersedes its headless-test status, but adds no desktop or offscreen-render result.

At a glance

- Shared chain is captured in DetectorSettings at Start; retained results use acquired settings, and the chain is locked while data exist.
- GAGG / NaI / LYSO / CsI / BGO map explicitly to transport materials. GAGG remains the default; the others are what-if comparisons. CsI:Tl uses CsI host transport with the activator omitted.
- Default scope uses real acquired times, 10 µs with 20% pretrigger; rate study explicitly re-spaces retained deposits.
- Realistic response is applied once before ADC simulation; ideal removes smear/noise/rise while retaining gain/tail/filter.
- CR-RC uses Q12 state and Q16 coefficients, with per-preset order 4 and explicit T_sum simulation shaping times; plotted values retain fractional codes.
- Headless engine/service/Core tests have passed; actual C#/Python/RTL comparisons and measured response are recorded below. Desktop and offscreen render verification were not run for this change.
- Default headless tests verify transport mapping (incl. CsI), shared response and acquisition locks.

Current normal-run totals and the distinction from retained desktop/RTL records are in VV.Studio. The inventories, CsI-only render limitation and CR-RC execution counts below are dated implementation records, not current solution totals.

Verification inventory

Area	Added coverage	Execution
Configuration and transport	Five supported-material rows; unknown preset rejection; GAGG default preserved; CsI captured by a real acquisition	Default headless tests
Shared response	GAGG and CsI across four preamps: waveform amplitudes equal the index-addressed measurement, Spectrum bins agree, Imaging uses the same response/windows; acquired identities/cache reset	Default headless tests
ADC waveform	Fixed-length raster compared to legacy without second intrinsic smear; actual retained MC associations; real-time/prehistory/pixel pooling; late timestamps; empty noise windows; ideal noise removal	Default headless tests
Local study/readout	Fixed-seed rate-study identity/immutability; matched trapezoid readout; suppression on overlap/saturation/partial windows; CR-RC caveat	Default headless tests

Area	Added coverage	Execution
Core requests/policies	Pure sample cap/conversion; prepared-pyramid identity; visible-only generation; held-event reuse; latest/next; acquired-chain policy; active-edit rejection; cancellation/latest-revision/new-run isolation; error reporting; Reset before selecting CsI for a new acquisition	Default headless tests
Evidence	Maximum window cost/cap; retained MC pulse readouts across supported parts/energies/modes; acquired mixed-field chain calibration and independent-service comparison	Opt-in, not run
Offscreen WPF	Real retained MC 10 µs / rate study / ideal, shared chain selector and pending/acquired labels, shared zoom/reset, both themes and sizes	Renderer edited, PNG generation not run

Source files: `tests/Gcam.Studio.Services.Tests/WaveformServiceTests.cs`, `AcquisitionServiceTests.cs`, `tests/Gcam.Studio.Tests/WaveformWorkspaceTests.cs`, `tests/Gcam.Studio.RenderTests/WaveformRenderTests.cs`; the latter is called within the existing single-STA/Application renderer. CsI adds five default waveform service cases, one real-acquisition case, one Core selector/reset case and twelve engine material cases. The existing opt-in render test now expects five selectable materials including CsI; its test-case count does not increase. The existing GAGG physics tests and scenario-loading test remain unchanged; only the material inventory count increases. RTL/cocotb files are unchanged.

CsI transport evidence is in `samples/materials/evidence/crystal_tables.json` and `samples/materials/CsI_checks.txt`. The evidence folder is outside the scenario test's non-recursive material JSON scan. The host uses 4.51 g/cm³; omitting the Tl activator follows the NaI:Tl convention, with no quantified dopant-specific transport error. At 661.7 keV, generated photo+incoherent attenuation is 0.0743482267 cm²/g, photo fraction 0.144938419 and linear attenuation 0.0335310502 /mm.

At 300/600/800 keV, adding independently generated coherent scattering gives like-total NIST comparisons. The absolute tolerances are 0.000074998 / 0.0000147103 / 0.0000129459 cm²/g: CsI's measured maximum relative residual 0.000105808471 rounded upward to 0.00011, plus each NIST and C# literal half-unit. Above 800 keV, tests assert the documented Klein–Nishina plus power-law extension and attenuation below NIST total. The measured extension errors against NIST photo+incoherent are +0.0699545 % at 1000 keV and −0.352789 % at 1250 keV; they are reported, not hidden in an acceptance band. `CsIMaterialTests` also checks both K edges, density/lookup, reference normalization, clamping and interpolation. C# attenuation/fraction tolerances follow each stored literal's last-place half-unit; the full-precision formula check uses 1e-14 for floating-point arithmetic only.

Reproduction

Run in the repository root. No desktop opt-in is needed or authorized by these commands.

```
dotnet build src/Gcam.Studio/Gcam.Studio.csproj -c Release -m:1 /nr:false /p:UseSharedCompilation=false
dotnet test tests/Gcam.Tests -c Release -m:1 /nr:false /p:UseSharedCompilation=false
dotnet test tests/Gcam.Studio.Tests -c Release -m:1 /nr:false /p:UseSharedCompilation=false
$waveSuite = [Diagnostics.Stopwatch]::StartNew()
dotnet test tests/Gcam.Studio.Services.Tests -c Release -m:1 /nr:false /p:UseSharedCompilation=false
$waveSuite.Stop()
"Services command wall time: $($waveSuite.Elapsed.TotalSeconds) s"
# Runner-reported test duration is distinct from command wall time/build overhead.
$env:GCAM_EVIDENCE_TESTS = '1'
dotnet test tests/Gcam.Studio.Services.Tests -c Release -m:1 /nr:false /p:UseSharedCompilation=false --
filter FullyQualifiedName~WaveformServiceTests --logger 'console;verbosity=detailed'
Remove-Item Env:GCAM_EVIDENCE_TESTS
$env:GCAM_RENDER_SNAPSHOTS = '1'
dotnet test tests/Gcam.Studio.RenderTests -c Release -m:1 /nr:false /p:UseSharedCompilation=false --logger
'console;verbosity=detailed'
Remove-Item Env:GCAM_RENDER_SNAPSHOTS
```

Planned outputs: `docs/assets/studio-render/waveform-{real10us|rate-study|ideal}-{dark|light}-{1280x800|1440x900}.png` (12 images). No PNGs were generated during the CsI validation. The renderer constructs/detaches the production XAML tree, measures/arranges and uses RenderTargetBitmap; it never calls Show, Studio App startup, creates an HWND or sends input. Shared chain selectors are included in each full-shell image and remain locked while acquired data exist. The actual MC fixture uses default scene/optics/detector, seed 12345, first 128 accepted ListModeSource events; trigger index 64. Rate study uses 1000 kcps, fixed seed 555, preserving original acquisition times in the event records.

Known limits and execution record

The ADC simulation adds analog/ADC noise after the analytic measurement. Shaped heights therefore are not the analytic MCA spectrum; recovered trapezoid energy is a scope estimate only. CR-RC recovered energy remains unavailable because trigger, phase and overlap estimation have not been validated; fractional state removes the whole-code accumulator deadband but is not an energy-estimator validation. No new resolution/pile-up precision test uses an invented tolerance. The small real-MC regression asserts deposit/time/pixel association, and the rate-study test asserts multiple pulses under explicitly simulated arrivals, not a promised real-scene overlap fraction.

Array-wide energy grouping follows the instrument's shared readout. The four ADC position channels, centroid positioning and pile-up mispositioning are absent. Background events follow the existing acquired stream. No afterglow or hand-added continuum is introduced.

Warm-up covers amplitude-dependent pulse support, eight tail constants and eight times the configured sum of nominal RC times but resets the local integer filter; it does not reproduce an acquisition-wide baseline state. CR-RC raw Q12 output is divided by 4096 into fractional output-code equivalents for display, without BLR. A partial acquisition window includes simulated baseline beyond live time and is labelled; its trapezoid energy is suppressed when the required readout horizon is unacquired. A capped request reserves warm-up inside the 10 M total raster samples, so its displayed length is slightly below 80 ms. Plot pyramids are built on the worker; integer shaper and pyramid calls are bounded but cancellation is checked around them rather than inside every iteration. Obsolete results cannot publish.

The first default parallel `dotnet build src/Gcam.Studio -c Release --no-restore` returned exit 1 without compiler errors. A sequential Studio build was then started and completed with **0 warnings /0 errors** (2m15.50s including compiler-server connection delays). While it ran, subsequent read-only PowerShell creation failed with `CreateProcessAsUserW failed: 5 (access denied)`. Per the requested fallback, no further verification commands were launched; the previously running build was only collected. Later tests, render paths, documentation and small source fixes had no final build/test result in that original execution. The subsequent headless execution below supplies current build/test evidence; visual snapshot inspection remains unexecuted for this change.

CR-RC arithmetic and response evidence

Implemented recurrence: coefficients remain Q16; input is sign-extended and shifted by F=12; pole-zero and each same-sample RC update retain fractional state. Raw output is divided by 4096.0 for plotted codes. Existing C#/Python calls default to F=0 and retain the legacy golden 304-code peak / sample-35 value 23. All CR-RC presets have order 4 with explicit T_sum 100/200/500 ns, K=20972/10486/4194. These are simulation times under T_sum=order×nominal Euler RC time, not peaking times, hardware evidence or equivalent DCR windows. The DCR integration fields remain independent. CR-RC energy recovery is still unavailable.

The bounded arithmetic domain is signed-16 samples, A∈[0,65536], K∈[1,65536], order 1–16, F 0–12. For B=2^(15+F), |imp|≤2B+1, every RC state remains in that interval by convexity of the floored update, and |u−acc|≤4B+2. Thus the maximum product for F=12 is **35,184,372,219,904 < 2^46**. Signed **48-bit** RTL states/products and C# long are sufficient for any input sequence in this domain, from reset, including valid holds. RTL enforces WACC≥WIN+F+Q+3; C#/Python reject inputs/parameters outside the bounded domain. This is an arithmetic bound, not physical hardware or Fmax validation.

Executed checks

Check	Result, 2026-10-02
`dotnet build Gcam.sln -c Release -m:1 /nr:false /p:UseSharedCompilation=false`	Success; 0 errors, 2 existing xUnit analyzer warnings in unchanged assertions (`ImagingServiceTests`, `WaveformServiceTests`)
`dotnet test Gcam.sln -c Release --no-build -m:1 /nr:false`	Engine 272 passed; Studio Core 164 passed; Services 75 passed / 7 Evidence skipped; UI oracle 13 passed / 14 desktop skipped; renderer 1 skipped. **524 passed / 22 skipped / 0 failed**. No desktop or renderer opt-in enabled.
CR-RC new .NET coverage	10 engine cases (metadata independence, low-energy retention with derived arithmetic bound, signed full-scale bounds, parameter rejection, vector export); 3 Services cases for fractional plotted codes/time labels/readout suppression
Icarus 12.0 / cocotb 2.0.1 / python.org 3.13	**137 passed / 0 failed / 0 skipped**: 26 configurations × 5 CR-RC cases =130; 6 unchanged trapezoid and 1 BLR case. Supported GAGG/NaI/LYSO/BGO × three presets × F=0/12, plus legacy F=0/12.
C# / Python / actual RTL equality	**26 vectors, 452,608 output samples, zero mismatches**. Each compares independently pinned A/K/order/F/Q/width; finite-rise energies, threshold sweep, negative/noisy/full-scale histories, valid gaps, reset and exact accepting-edge output offset 0.
Preserved tolerance	Legacy 400→800 peak ratio still requires abs(ratio−2)<0.05, unchanged. This benchmark tolerance is not extended to whole-code low-energy preset paths; those use exact sample equality.
Derived precision tolerance	For the floating recurrence on identical rounded ADC samples and A/K, each RC update floor contributes <1 raw unit with accumulated error ≤1/K; deconvolution contributes <1. Maximum difference is bounded by **(1+order×65536/K)/4096 output codes**. All-sample tests use that bound, not a tuned keV tolerance.
Measurement artifact build	`dotnet build docs/assets/crrc-presets/Measure.csproj -c Release`: success, 0 warnings / 0 errors

The prior source-inventory counts above are not current totals. Default tests pass; opt-in maximum-window, cross-parts MC calibration, render and desktop evidence remains unexecuted in this change. The full RTL suite passed 137 cases; after retaining an explicit finite-rise piled-pulse stimulus, the final CR-RC-only matrix was rerun and passed all 130 cases with no failures/skips. The seven unchanged trapezoid/BLR cases retain their full-suite result.

Response remeasurement

The standalone measurement source calls the **implemented** `Waveform.CrrcInt` for all paths, including legacy F=0, shared-K Q12 and configured Q12. Complete data: amplitude.csv, resolution.csv. No data were copied from a floating candidate and presented as implemented output.

Conditions: GAGG / S13360-3050 / AD9648, 125 MSPS; gain 4.0955 codes/keV, ENOB noise 1.32746148 codes RMS. Finite input rise 90 ns; CSP tails 150/320/800 ns; matched A=62132/63918/64884. Each deterministic pulse is isolated at sample 100 in 1024 samples, no ENOB/analog noise or amplitude smear. This finite-rise measurement stimulus is distinct from Studio's instantaneous-rise ideal mode.

Entries below are peak output-code equivalents at **32 / 122 / 662 keV**:

Preset	Legacy shared K, F=0	Shared K, Q12	Implemented preset K, Q12
Fast	0 / 5 / 47	2.505127 / 9.609375 / 51.970459	**2.234131 / 8.518555 / 46.140381**

Preset	Legacy shared K, F=0	Shared K, Q12	Implemented preset K, Q12
Original	0 / 14 / 90	4.591064 / 17.458740 / 94.780518	**2.625732 / 10.028809 / 54.438232**
Slow	0 / 17 / 114	5.719238 / 21.741943 / 117.933350	**1.567627 / 5.989258 / 32.526855**

At **32.1 keV**, implemented Q12 maxima are **2.234131 / 2.634521 / 1.569580**. The inherited seed-1 ENOB-on stimulus still gives F=0 maxima **−1/−1/0** at 32.1 keV (0/0/1 with the old generator's noise) and **47/90/114** at 662 keV; removing ENOB noise gives 0/0/0 at 32.1 keV. Q12 output fractions, not an extra plotted gain, retain the signal. Configured floating peak times are **104/200/448 ns**. Nominal stage times are **25/50/125 ns**; the difference reinforces why T_sum cannot be labelled peaking time.

Resolution repeats **6000 isolated traces** at each energy/mode/path. Sample noise seeds 1…6000, amplitude RNG seed 901. Electronics = 3-keV-RMS analog sample noise plus ADC ENOB; full adds one `FrontEndModel.Measure` amplitude draw before rasterising. Baseline = mean of 64 pretrigger shaped samples. Read amplitude at the known noiseless floating peak time. For g_float=unquantised-input floating 662-keV peak/662, report R_equiv=2.3548×sample SD/(g_float×E). These are Gaussian-equivalent widths, not fitted FWHMs of the discrete/biased legacy distributions and not an implemented energy estimator.

Entries are percentages at **32 / 122 / 662 keV**:

Preset/path	Electronics R_equiv %	Full R_equiv %	Full mean bias %
Fast legacy F=0	85.344 / 22.558 / 4.570	86.271 / 23.848 / 6.456	−10.069 / −5.942 / −1.401
Original legacy F=0	47.733 / 12.810 / 2.507	49.275 / 14.880 / 5.161	−7.281 / −3.830 / −0.869
Slow legacy F=0	37.974 / 10.367 / 1.919	39.776 / 12.904 / 4.794	−4.959 / −3.210 / −0.655
Fast shared-K Q12	78.290 / 20.536 / 3.786	79.531 / 21.905 / 5.874	−0.265 / −0.055 / −0.001
Original shared-K Q12	42.605 / 11.178 / 2.060	44.739 / 13.498 / 4.929	−0.106 / −0.017 / +0.005
Slow shared-K Q12	34.297 / 8.997 / 1.659	36.931 / 11.748 / 4.774	−0.068 / −0.005 / +0.007
Fast implemented	**59.141 / 15.512 / 2.859**	**60.714 / 17.261 / 5.312**	−0.060 / −0.003 / +0.008
Original implemented	**16.507 / 4.330 / 0.798**	**21.448 / 8.724 / 4.551**	+0.228 / +0.070 / +0.022
Slow implemented	**7.679 / 2.013 / 0.371**	**15.658 / 7.816 / 4.492**	−0.048 / −0.001 / +0.008

Re-measured 2026-10-02 after the random generator was replaced (the trace noise seeds 1…6000 and the amplitude stream 901 now come from xoshiro256**; the old generator's consecutive seeds gave shifted copies of one noise sequence). Earlier implemented-path values (old generator): electronics 59.071 / 15.497 / 2.855, 16.176 / 4.242 / 0.782, 7.697 / 2.019 / 0.372; full 60.781 / 17.381 / 5.456, 21.317 / 8.848 / 4.693, 16.083 / 8.079 / 4.653 (Fast / Original / Slow, 32 / 122 / 662 keV). The changes (up to −0.16 points at 662 keV full, +0.33 at 32 keV electronics) are one realisation against another: a 6000-trace width has a relative standard error of ≈ 0.9 %, and all presets share the same noise and amplitude draws, so their changes are correlated. This is reproducibility evidence, **not** an adopted energy accuracy tolerance. The physical analytic chain alone predicts approximately **13.81/7.686/4.569%**; the additional waveform sample noise explains the larger measured widths. The predicted analytic number has not been substituted for measured waveform resolution.

Configured full 32-keV width ranges over three disjoint 2000-trial batches are Fast **59.90–61.69%**, Original **20.96–21.98%**, Slow **15.26–15.89%**; at 662 keV **5.233–5.368 / 4.413–4.634 / 4.363–4.562%** (old generator: 60.02–62.15 / 21.20–21.38 / 15.90–16.22%; 5.418–5.513 / 4.672–4.719 / 4.619–4.694%). These are observed seed-batch spreads, not confidence intervals or pass thresholds. Q12 quantisation granularity in calibrated output is **0.00350/0.00297/0.00497 keV**; it is not detector energy resolution. 27…37 keV in 0.01-keV increments yields configured **68/303/674 distinct peak levels**, all nonzero, versus legacy **1/1/3 levels** and **100/100/56.54%** zero maxima. Input ADC rounding still causes plateaus.

Reproduction: run the supplied console project from a fresh `%TEMP%/gcam-*` working directory, passing the absolute path to `docs/assets/crrc-presets/Measure.csproj` to `dotnet run -c Release --project ...`. It writes `deterministic.csv`, `resolution.csv` and prints coefficient/staircase/reproduction summaries in that scratch directory. Its three paths are `current`, `currentQ12`, `configured`; output CSV columns retain the estimator/gain definitions above, plus gated-peak diagnostics and observed batch spread. The gated peak is a diagnostic only: selecting maxima can reduce width while biasing the mean upward.

Not established here: threshold-trigger efficiency, arrival phase / variable-rise response, pile-up energy recovery, full-scale analog dynamic range calibration, synthesis/device Fmax, physical hardware topology, desktop label layout or new offscreen renders. Four-channel charge division and hardware validation remain separate. All coefficients/state tolerances were kept or derived; none was loosened to obtain a pass.

On 2026-10-02, the final sequential Release solution build passed with 0 errors and two existing xUnit analyzer warnings (xUnit2000 in ImagingServiceTests and xUnit2012 in WaveformServiceTests). `dotnet test Gcam.sln -c Release --no-build --no-restore -m:1 /nr:false` passed after that build. All three opt-in environment variables (numerical evidence, rendering and desktop) were unset.

Suite	Passed	Skipped	Failed
Engine	274	0	0
Studio Core	165	0	0
Studio Services	78	7	0
UI oracle / desktop assembly	13	14	0
Offscreen render assembly	0	1	0
Total	530	22	0

The service skips are long numerical evidence; the UI/render skips execute no desktop or WPF rendering. Only the 13 headless oracles ran in the UI assembly. The full console output is retained in `samples/materials/CsI_validation.txt`; the Python table checks passed separately. Engine cases increase 262→274, Core 164→165 and Services 72→78 passed (seven evidence skips remain). Long numerical evidence, offscreen renders and desktop input remain independently opt-in; they are not covered by the default suite's pass result. RTL was not rerun because transport-data additions change no RTL, native shaper or waveform contract.